Beyond Recall: A Fine-Grained Multimodal Benchmark for Agent Memory across Images, Videos, and Documents

Chenyu Wang*, Junkai Chen*, Zhongkai Xue, Zhetao Guo, Shu Wu†

Institute of Automation, Chinese Academy of Sciences
University of Chinese Academy of Sciences
Beijing Zhongguancun Academy
chenyuw494@gmail.com, shu.wu@nlpr.ia.ac.cn

Abstract

Existing agent-memory benchmarks test factual recall from text, while multimodal benchmarks place visual evidence in a single turn or short context. They rarely distribute multimodal evidence across sessions occurring at different times, and their broad task categories make it difficult to evaluate specific memory abilities. We introduce *Beyond Recall*, a fine-grained benchmark for long-term agent memory containing 1,865 unique images, 85 multi-page PDF documents, and 314 videos. We organize the benchmark into 11 tasks across three dimensions: retrieval, reasoning, and management. Its questions require task-relevant multimodal evidence, reducing the chance of answering them from text alone. We construct the benchmark through a human-in-the-loop pipeline. Human experts create evidence, distractors, questions, and answers from real-world multimodal materials, while language models summarize topics and adapt context from open-source datasets into multi-session conversations. Human reviewers then check the resulting examples for evidence grounding, consistency, ambiguity, and answer leakage. *Beyond Recall* contains 1,163 long-term conversation trajectories, 1,269 question–answer pairs, and 23,775 timestamped sessions. We evaluate agent-memory frameworks, retrieval-augmented pipelines, and full-context baselines under a unified protocol. Performance varies greatly across memory abilities, and no single method performs best on all tasks. Across the evaluated retrieval and memory systems, complete evidence access averages 70.0%, yet average answer accuracy on these cases is only 40.7%. Together with our error analysis, this gap shows that current systems still struggle to use multimodal evidence effectively and combine information across sessions. Project website: https://beyond-recall.github.io/.

Introduction

Large language model agents are evolving from short-term conversational systems into long-term assistants, as reflected by widely used assistant products such as ChatGPT and Microsoft Copilot (OpenAI 2022; Mehdi 2023) and by research agents that sustain interaction and task execution over extended periods (Park et al. 2023; Wang et al. 2023; Yao et al. 2023). To handle increasingly complex tasks and multimodal information (images, videos, etc.), a growing number of multimodal agent systems have emerged (Alayrac et al. 2022; Yang et al. 2023; Liu et al. 2024c). Meanwhile, agent-memory frameworks have been developed to help agents store, retrieve, organize, and update information from past interactions (Shinn et al. 2023; Packer et al. 2023; Zhong et al. 2024; Chhikara et al. 2025; Xu et al. 2025). Therefore, systematically evaluating the memory capabilities of these frameworks has become increasingly important.

However, current agent-memory systems are still commonly evaluated on text-centered benchmarks, while existing multimodal evaluations usually focus on a single type of visual evidence. For example, LoCoMo evaluates long-term conversations through question answering, summarization, and multimodal dialogue generation (Maharana et al. 2024). LongMemEval evaluates information extraction, multi-session and temporal reasoning, knowledge updates, and abstention in textual histories (Wu et al. 2025). MemoryAgentBench studies incremental text interactions and selective forgetting (Hu, Wang, and McAuley 2025). Recent benchmarks such as Mem-Gallery and MemLens extend long-term memory evaluation to multi-session conversations containing images (Bei et al. 2026; Ren et al. 2026), while document and video benchmarks emphasize long visual inputs without explicit agent-memory construction (Ma et al. 2024; Chandrasegaran et al. 2024). Although these benchmarks have advanced long-context and multimodal memory evaluation, they usually cover only a limited subset of modalities and thus cannot comprehensively assess whether a single memory system can retrieve, integrate, and manage heterogeneous evidence across long-term sessions under a unified, fine-grained framework.

To address these limitations, we introduce *Beyond Recall*, a fine-grained benchmark for evaluating long-term multimodal memory in agents. As illustrated in Figure 1, we organize agent memory into three dimensions: retrieval, reasoning, and management, covering 11 fine-grained tasks and 1,269 question–answer pairs. To systematically evaluate heterogeneous multimodal information within a unified framework, *Beyond Recall* includes 1,865 images, 85 multi-page PDF documents, and 314 videos, as summarized in Table 1. To ensure multimodal information is essential for answering, each question is grounded in task-relevant multimodal evidence, requiring systems to extract and use information from the associated media rather than rely on text alone. To accelerate benchmark construction without compromising data quality,

*These authors contributed equally and share first authorship.
†Corresponding author.

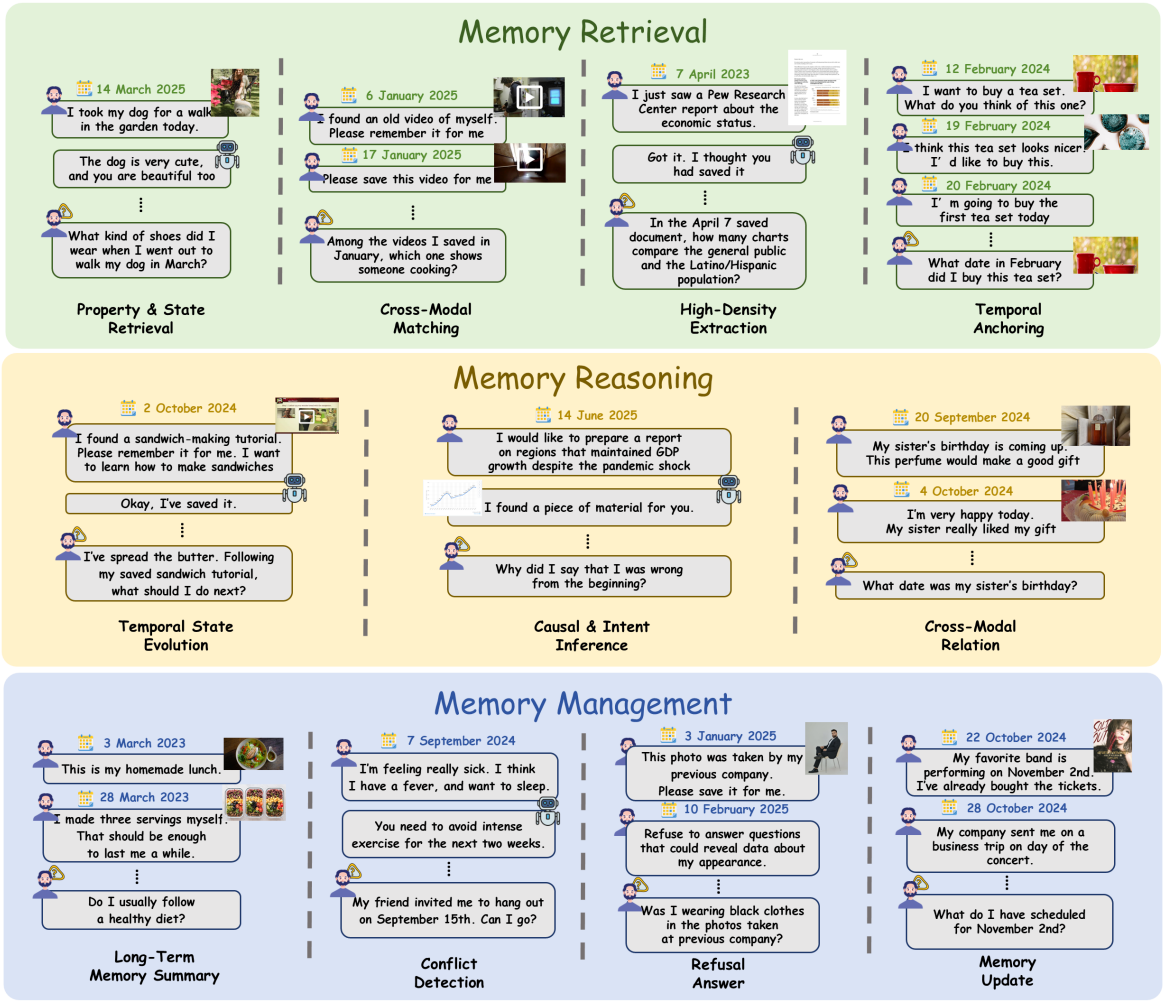

Figure 1: Task organization of Beyond Recall. Each task shows key evidence fragments and a corresponding question.

we design a human-in-the-loop pipeline in which language models assist experts in developing data examples.

We conduct a unified evaluation of representative agent-memory frameworks, retrieval-augmented pipelines, and full-context baselines under identical settings. The results reveal substantial performance differences across retrieval, reasoning, and management tasks, with no single method consistently performing best across all capabilities. More importantly, across the evaluated retrieval and memory systems, complete evidence access averages 70.0%, yet average answer accuracy on these cases is only 40.7%. Together with our experimental analysis, this gap demonstrates that successful retrieval alone is insufficient and that effective multimodal evidence use and cross-session information integration remain major challenges for current memory systems.

Our main contributions are threefold:

- We introduce *Beyond Recall*, a multimodal benchmark covering images, charts, multi-page PDF documents, and videos, organized into 11 tasks spanning retrieval, reasoning, and management. Moreover, our evaluation results reveal distinct capability profiles across models, with no single method performing best on every task.
- We design a human-in-the-loop data construction pipeline in which language models provide storyline ideas and human experts reconstruct reliable fine-grained details, evidence, questions, and answers from real media. These questions strongly depend on visual information and require integrating multiple evidence cues.
- We find that multimodal use remains the central bottleneck. Our experiments show a clear gap between retrieving evidence and answering questions correctly: even when relevant evidence is successfully retrieved, current systems often fail to extract and effectively use it.

Related Work

Long-Term Memory Evaluation for Agents. Large-scale language model agents increasingly rely on external memory mechanisms to maintain and retrieve information beyond individual interactions (Park et al. 2023; Packer et al. 2023; Shinn et al. 2023). Recent benchmarks evaluate increasingly

Benchmark	QA pairs	Context Scale	Image	PDF	Chart	Video	Memory taxonomy
LoCoMo (Maharana et al. 2024)	1,986	9.2K	√	×	×	×	5
LongMemEval (Wu et al. 2025)	500	115K–1.5M†	×	×	×	×	5
MMDU (Liu et al. 2024d)	1,645	8.2K	√	×	×	×	–
Mem-Gallery (Bei et al. 2026)	1,711	–	√	×	×	×	9
MemLens (Ren et al. 2026)	789	32K–256K†	√	×	×	×	9
MMLongBench-Doc (Ma et al. 2024)	1,062	21.0K	√	√	√	×	–
HourVideo (Chandrasegaran et al. 2024)	12,976	–	×	×	×	√	–
Beyond Recall	**1,269**	**10.4K**	√	√	√	√	**11**

Table 1: Comparison of *Beyond Recall* with representative long-term memory and multimodal benchmarks. The table summarizes QA-pair counts, context scale, modality coverage, and memory-taxonomy size. "Context Scale" reports average context length when available; † marks a range, controlled budget, or estimate. "Memory taxonomy" reports the number of fine-grained categories or tasks when defined; "–" indicates that a statistic is unavailable or inapplicable.

complex multi-session settings. LoCoMo covers question answering, summarization, and multimodal dialogue generation in long conversations (Maharana et al. 2024), while LongMemEval decomposes memory into extraction, multi-session and temporal reasoning, updates, and abstention (Wu et al. 2025). MemoryAgentBench further evaluates memory through incremental interactions and controlled forgetting (Hu, Wang, and McAuley 2025). However, these benchmarks primarily study text-based memory and do not fully capture agents that preserve, retrieve, and reason over heterogeneous multimodal experiences accumulated over time. Mem-Gallery and MemLens extend evaluation to image-containing interactions (Bei et al. 2026; Ren et al. 2026). Nevertheless, they remain limited in modality coverage or capability granularity, leaving fine-grained evaluation across images, videos, and documents underexplored.

Multimodal Evidence Reasoning. Multimodal benchmarks study evidence extraction and reasoning across images, documents, charts, and videos (Antol et al. 2015; Mathew, Karatzas, and Jawahar 2021; Masry et al. 2022; Li et al. 2024). MMDU covers multi-turn dialogue with multiple images (Liu et al. 2024d), MMLongBench-Doc studies long-context multimodal documents (Ma et al. 2024), and HourVideo targets temporal reasoning over long videos (Chandrasegaran et al. 2024). They do not primarily test evidence retained and revised across separated sessions.

Memory Architectures for LLM Agents. A variety of memory mechanisms have been proposed to extend LLM agents beyond limited context windows for persistent and personalized interaction (Park et al. 2023; Packer et al. 2023). MemoryBank models long-term conversational memory through storage and updating mechanisms (Zhong et al. 2024). Mem0 extracts salient memories and consolidates them for future retrieval (Chhikara et al. 2025), while A-Mem organizes memories as linked evolving notes inspired by the Zettelkasten framework (Xu et al. 2025). MIRIX further introduces specialized memory structures for multimodal agents (Wang and Chen 2025).

In parallel, retrieval-augmented generation and long-context modeling provide alternative strategies for accessing historical information. These systems differ substantially in memory representation, update policy, and retrieval strategy, making it difficult to compare them without a unified evaluation framework. *Beyond Recall* provides such a framework by systematically evaluating end-to-end memory behavior across retrieval, reasoning, and management capabilities.

RAG combines generation with retrieved evidence (Lewis et al. 2020), often relying on dense retrieval or hybrid lexical-dense retrieval to select relevant context (Karpukhin et al. 2020; Khattab and Zaharia 2020; Robertson et al. 1994). Multimodal variants retrieve and combine images and text (Chen et al. 2022). Full-context models provide a simple alternative but may underuse information in long inputs (Liu et al. 2024b). We include dense, hybrid, caption, and native-multimodal baselines to test the value of explicit memory and raw media under a unified evaluation protocol.

Beyond Recall

Problem Formulation

An instance is $x = (H, q, a, E)$, where $H = (S_1, \ldots, S_n)$ is an ordered sequence of sessions. Each session $S_i = (t_i, U_i)$ contains a timestamp t_i and utterances U_i, each consisting of a speaker, text, and optional media. After the history, the target user issues a query q containing text and optional media. Here, a is the reference answer, and E is the minimal set of supporting utterances and media objects in H. Evidence identifiers are used only for validation and analysis and are hidden from evaluated systems.

Given a memory procedure M, the evaluation has two stages:

$$Z = M_{\text{write}}(H), \qquad \hat{a} = M_{\text{answer}}(q, Z), \tag{1}$$

where Z may be extracted facts, linked notes, a retrieval index, or the full history. This exposes errors from media perception through downstream reasoning.

Dataset Overview

Beyond Recall contains 1,269 question–answer pairs organized into three capability dimensions and 11 fine-grained tasks, as summarized in Figure 1. The retrieval dimension,

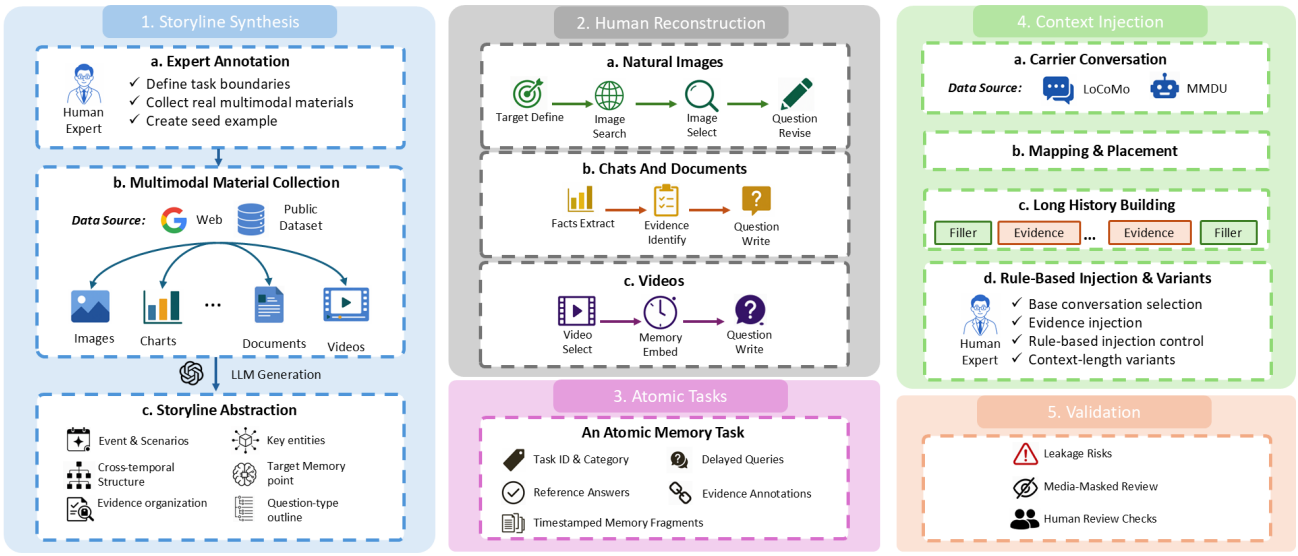

Figure 2: Construction pipeline of *Beyond Recall*. Human experts define capability boundaries and collect real multimodal materials. Language models assist storyline abstraction, after which experts reconstruct fine-grained, media-grounded memory fragments, questions, answers, and evidence annotations. Atomic tasks are injected into long carrier conversations from LoCoMo and MMDU, then reviewed for grounding, consistency, ambiguity, and answer leakage.

which focuses on locating detailed information across modalities and time, contains 493 questions across Property & State Retrieval (139), High-Density Extraction (123), Cross-Modal Matching (116), and Temporal Anchoring (115). The reasoning dimension, which integrates multi-source evidence to infer states, causes, intentions, and relations, contains 370 questions across Temporal State Evolution (125), Causal & Intent Inference (121), and Cross-Modal Relation (124). The management dimension, which summarizes, verifies, and updates long-term information while avoiding unsupported answers, contains 406 questions across Long-Term Memory Summary (71), Conflict Detection (71), Refusal Answer (108), and Memory Update (156).

Data Construction

As illustrated in Figure 2, Beyond Recall combines expert-seeded storyline synthesis, model-assisted scenario ideation, and human reconstruction grounded in real media. Appendix Figure A1 provides a detailed view of the expert-to-model-to-expert transition used in this process.

Storyline Synthesis Human experts define each task's capability boundary, collect real multimodal materials, and author seed examples with timestamped fragments, delayed questions, reference answers, and evidence relations.

Language models use the task definitions, seed examples, and materials to explore events, entities, and long-term story structures. Only the resulting high-level storylines are retained; generated fragments, questions, answers, and evidence annotations are discarded during reconstruction.

Human Reconstruction After storyline abstraction, human experts reconstruct the final samples from scratch using the original multimodal materials. This step is central to the reliability of the benchmark. Experts rewrite fine-grained memory fragments, assign timestamps, define temporal or causal relations, annotate evidence, and create the final questions and reference answers. The resulting samples are therefore grounded in real media and expert judgment, rather than being lightly edited versions of model-generated drafts.

Atomic Tasks The reconstructed samples are represented as atomic memory tasks containing identifiers, categories, timestamped fragments, delayed queries, reference answers, and evidence annotations. This layer enables rigorous validation before distractors are introduced; queries hide internal identifiers, multi-evidence answers must be fully supported, and refusal cases contain no sufficient evidence.

Context Injection To evaluate memory in long histories, we inject atomic tasks into multi-session carrier conversations from LoCoMo (Maharana et al. 2024) and MMDU (Liu et al. 2024d), using them as distractor context rather than answer sources (Bai et al. 2024; Liu et al. 2024b).

A fixed-seed conversation is selected and evidence fragments are mapped to its primary speaker. Deterministic scripts insert timestamped evidence sessions, add 4–7 filler sessions, preserve temporal order, and withhold evidence labels. The benchmark averages 10.4K tokens; 4K, 8K, and 12K variants retain all evidence while pruning filler sessions, with results reported in the appendices.

Validation Automatic checks and media-masked human review verify grounding, consistency, uniqueness, ambiguity, leakage, distractors, updates, conflicts, and refusals. Reviewers also explicitly confirm that each answer is uniquely

recoverable from annotated evidence and that temporal relationships remain coherent across injected sessions.

Experimental Setup

Backbone Models

For answer generation, we evaluate three backbone models of different sizes: `Qwen3.5-9B`, `Qwen3.5-27B`, and `Qwen3.5-35B-A3B` (Qwen Team 2026). Each model generates answers from full context, retrieved sessions, or explicit memories and supports native multimodal input. The main results in Table 2 use `Qwen3.5-9B`. For memory extraction and evolution, we use `Qwen3.5-35B-A3B` across all agent-memory frameworks to ensure a consistent comparison. For configurations that cannot directly process images or videos, we use `Qwen3.5-35B-A3B` as a shared captioning model to convert visual content into cached textual descriptions. For automatic answer evaluation, we use `Qwen3.5-122B-A10B` as a shared LLM judge for all configurations under the same evaluation rubric.

Compared Configurations

We group the evaluated methods into three configurations according to how historical information is made available at query time: Full Context, Retrieval, and Explicit Memory.

Full context. **Full-Context Multimodal** sends the full context and task-relevant media directly to the answering model as native multimodal input, following recent progress in vision-language foundation models (Alayrac et al. 2022; Liu et al. 2024a). **Full-Context Caption** first converts multimodal data into textual captions and then sends the caption-augmented full context to the answering model.

Retrieval. **RAG-Dense** follows the retrieval-augmented generation paradigm (Lewis et al. 2020), embedding and retrieving relevant sessions with BGE-small-en-v1.5 (Xiao et al. 2024) before supplying the top-ranked sessions to the answer model. **RAG-Hybrid** extends this configuration by combining BGE dense retrieval with BM25 lexical retrieval (Robertson et al. 1994) and merging their results using reciprocal rank fusion (Cormack, Clarke, and Buettcher 2009). **MuRAG** (Chen et al. 2022) jointly retrieves relevant textual sessions and visual evidence, then provides the answer model with the retrieved sessions and their associated images or videos for multimodal question answering tasks.

Explicit memory. **Mem0** (Chhikara et al. 2025) extracts and consolidates session-level information into structured persistent memories, then retrieves relevant memories for question answering. We evaluate two Mem0 variants: **Mem0-A**, the legacy non-graph framework, and **Mem0-O**, the newer framework. **A-Mem** (Xu et al. 2025) represents sessions as interconnected notes that are dynamically linked and evolved before retrieval. **MIRIX** (Wang and Chen 2025) uses specialized agents to organize multimodal information into six memory types and retrieves task-relevant memories for answering each query. These methods differ in how they select, consolidate, update, and connect information across sessions. Comparing them under backbones and prompts allows us to separate framework-level memory behavior from differences in answer generation, while revealing whether persistence improves access to heterogeneous evidence over conventional retrieval and full-context baselines.

Protocol and Metrics

We evaluate all configurations on the same 1,269 questions using trajectory-specific memory stores or retrieval indices and chronologically ordered sessions under identical conditions. Systems receive the history and query, but not reference answers, task labels, evidence annotations, or source identifiers. Caption-based systems share cached descriptions, whereas native multimodal systems receive the source media. All configurations use the same answering prompt and backbone model, and execution or parsing failures are counted as incorrect. A shared temperature-zero LLM judge evaluates semantic correctness and is validated against human judgments on 150 predictions. Our primary metric is strict micro-averaged accuracy, $\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} y_i$, reported by capability dimension, task, and evidence modality.

Let E_i be the annotated evidence units for question i and R_i^K the top-K units returned by a system. We set $K = 10$ for all retrieval and memory systems and report AnyHit@K, Complete Evidence-Source Access (CESA@K), and accuracy conditioned on complete access:

$$\mathrm{AnyHit@K} = \frac{1}{N}\sum_{i} \mathbf{1}(E_i \cap R_i^K \neq \emptyset), \tag{2}$$

$$\mathrm{CESA@K} = \frac{1}{N}\sum_{i} \mathbf{1}(E_i \subseteq R_i^K), \tag{3}$$

$$\mathrm{Acc} \mid \mathrm{CESA} = \frac{\sum_i y_i c_i}{\sum_i c_i}, \quad c_i = \mathbf{1}(E_i \subseteq R_i^K). \tag{4}$$

RAG-Dense, RAG-Hybrid, and A-Mem preserve exact turn-level provenance; Mem0 and MIRIX use timestamp-based source-session proxies, while full-context methods measure evidence exposure rather than retrieval recall.

Results and Analysis

Overall Results and Capability Breakdown

Table 2 presents the full-dataset evaluation of all nine configurations under identical settings on *Beyond Recall*. We find that: (1) no single method consistently leads across all tasks; (2) native multimodal input still outperforms caption-based representations; (3) current memory frameworks remain limited in organizing information at a global level.

On retrieval tasks, the performance gaps are relatively small, suggesting that existing systems already have some ability to locate relevant information. Reasoning tasks show a clearer need for multimodal evidence. Native multimodal configurations achieve an average reasoning accuracy of 32.52%, compared with 26.29% for caption or text based configurations. This gap indicates that raw images, videos, and documents provide useful additional evidence for reasoning. On management tasks, full-context methods perform strongly, with Full-Context Multimodal reaching 50.74% and Full-Context Caption reaching 42.61%. This pattern suggests

Method	Memory Retrieval					Memory Reasoning				Memory Management					All
	PSR	HDE	CMM	TA	Ret.	TSE	CII	CMR	Reas.	LTMS	CD	RA	MU	Mgmt.	
MuRAG	53.2	49.6	42.2	**66.1**	**52.7**	37.6	**48.8**	37.9	**41.4**	36.6	**49.3**	25.9	49.4	40.9	**45.6**
Full-MM	43.9	22.0	52.6	39.1	39.4	18.4	46.3	38.7	34.3	**49.3**	42.3	**63.9**	46.2	**50.7**	41.5
Mem0-A	**58.3**	22.8	**59.5**	36.5	44.6	**40.0**	32.2	29.8	34.1	45.1	38.0	23.2	49.4	39.7	39.9
Mem0-O	51.8	35.0	55.2	26.1	42.4	31.2	20.7	32.3	28.1	47.9	31.0	25.9	45.5	38.2	36.9
Full-Cap	50.4	36.6	46.6	25.2	40.2	21.6	22.3	36.3	26.8	36.6	12.7	41.7	**59.6**	42.6	37.0
A-Mem	52.5	**55.3**	39.7	36.5	46.5	20.0	22.3	**43.5**	28.6	32.4	9.9	40.7	52.6	38.4	38.7
MIRIX	35.3	19.5	53.4	31.3	34.7	21.6	18.2	25.8	21.9	**49.3**	25.4	25.0	35.3	33.3	30.5
RAG-D	53.2	52.0	36.2	35.7	44.8	17.6	15.7	37.9	23.8	32.4	4.2	26.9	46.8	31.5	34.4
RAG-H	41.7	48.8	36.2	27.0	38.7	8.8	10.7	29.0	16.2	33.8	7.0	31.5	44.2	32.5	30.2

Table 2: Strict accuracy (%) on the main *Beyond Recall* dataset (10.4K tokens on average) using the Qwen3.5-9B backbone. Ret., Reas., and Mgmt. denote capability averages. Full-MM/Full-Cap: full-context multimodal/caption; Mem0-A/Mem0-O: legacy non-graph/new framework; RAG-D/RAG-H: dense/hybrid. Best values are bolded.

Method	Any-hit	Complete	Overall	Acc. | complete
MuRAG	94.33%	73.21%	45.63%	52.21%
Full-MM	100.00%	98.55%	41.53%	41.11%
Mem0-A	97.32%	79.83%	39.87%	43.73%
Mem0-O	96.38%	68.87%	36.80%	41.99%
Full-Cap	100.00%	100.00%	37.04%	37.04%
A-Mem	99.61%	90.70%	38.70%	35.62%
MIRIX	44.02%	15.26%	30.50%	38.95%
RAG-D	99.45%	89.36%	34.44%	36.86%
RAG-H	95.04%	72.89%	30.18%	35.46%

Table 3: Method-level evidence access and conditional answer accuracy. Mem0 and MIRIX use session-level proxies; full-context rows measure protocol-level exposure rather than retrieval Recall@K.

Method	Image	Chart	PDF	Video
MuRAG	51.0	75.7	32.7	**33.8**
Full-MM	47.1	32.9	16.4	32.5
Mem0-A	**51.1**	38.6	14.5	14.2
Mem0-O	44.0	58.6	9.1	17.1
Full-Cap	41.7	55.7	23.6	24.2
A-Mem	41.5	**82.9**	**36.4**	20.8
MIRIX	39.4	10.0	12.7	7.9
RAG-D	38.2	80.0	30.9	14.6
RAG-H	32.5	75.7	30.9	13.8

Table 4: Strict accuracy (%) of each evaluated framework across image, chart, PDF, and video evidence. The best point estimate in each modality column is bolded.

that memory management still benefits from complete historical context, while current memory frameworks remain limited in global association and state organization.

Retrieval Hit Analysis

Table 3 reports how well each framework accesses key evidence and how accurately it answers under different evidence-access conditions. We observe two main patterns: (1) retrieving all annotated evidence remains important for downstream answering; (2) complete evidence access does not guarantee correct answers.

First, complete evidence access is more useful than partial evidence access. Across the seven retrieval and memory rows in Table 3, excluding the two full-context controls, average answer accuracy rises from 36.59% overall to 40.69% when all annotated evidence is accessed, a gain of 4.10 points. This indicates that partial evidence access is often insufficient: many questions require multiple fragments to be jointly available before the model can produce the correct answer. Second, high access rates still coexist with low answer accuracy. The same systems achieve 89.45% Any-hit/access and 70.02% Complete access on average, yet their average overall accuracy remains only 36.59%. Even under complete access, conditional accuracy is only 40.69%, showing that systems often retrieve the necessary evidence but fail to use it effectively for answer generation in practice.

Framework Capability Analysis

Table 4 shows that the evaluated frameworks have different capability profiles across evidence types, and no method performs best on all modalities within our evaluation. Mem0-A achieves the highest accuracy on images at 51.1%, A-Mem performs best on charts and PDFs at 82.9% and 36.4%, and MuRAG performs best on videos at 33.8%.

The modality-level results also reveal framework-specific limitations. A-Mem and the two text-oriented RAG baselines perform strongly on charts, but reach only 13.8–20.8% on videos. In contrast, MuRAG and Full-Context Multimodal achieve the highest video scores, consistent with preserving native temporal evidence. Mem0-A performs best on image questions, but reaches only 14.5% on PDFs and 14.2% on videos. By comparison, A-Mem preserves more chart and PDF information. These results show that effective agent memory requires not only retrieving relevant records, but also representations and update mechanisms that preserve

Input condition	Overall	Image	Chart	PDF	Video
Question-Only	6.15%	1.69%	0.00%	0.00%	8.33%
Text-Only	24.51%	29.83%	0.00%	9.09%	16.67%
Full-Context Caption	37.04%	41.67%	55.71%	23.64%	24.17%
Full-Context Multimodal	41.53%	47.10%	32.86%	16.36%	32.50%
Evidence-Only Caption	61.35%	69.85%	81.43%	24.24%	35.00%
Evidence-Only Multimodal	60.22%	68.69%	44.29%	24.24%	46.25%

Table 5: Strict accuracy under progressively richer evidence and oracle-evidence conditions. The modality columns contain questions annotated with the corresponding evidence type. Evidence-only rows directly provide annotated evidence and exclude unrelated history; their overall and PDF scores use cases with verifiable complete oracle evidence delivery, with corrected Oracle PDF handling.

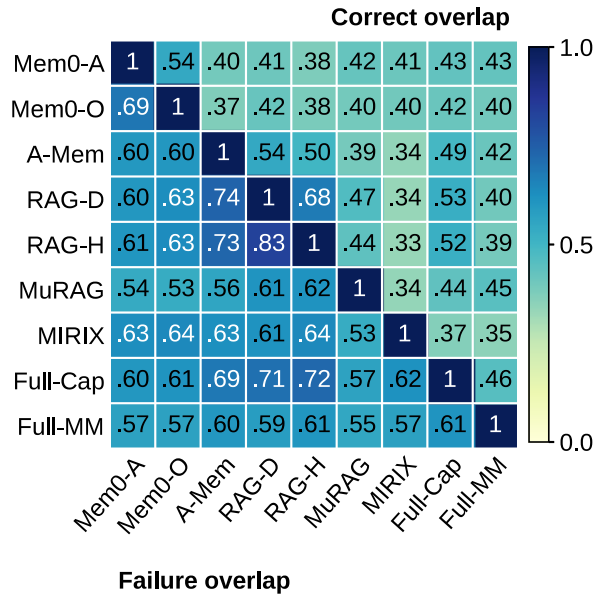

Figure 3: Pairwise Jaccard similarity across frameworks. The upper triangle shows overlap between correct-answer sets, and the lower triangle shows overlap between failure sets.

the evidence properties needed by each modality.

Figure 3 compares the question-level outcome distributions of the evaluated frameworks, showing both similarity and complementarity among them. Mem0-O and Mem0-A make the same decisions on 77.4% of the questions, while RAG-Dense and RAG-Hybrid are even more similar, with a decision agreement rate of 87.5%. At the same time, different frameworks also show complementarity. MuRAG and Mem0-A jointly answer 323 questions correctly, while MuRAG uniquely answers 256 questions correctly and Mem0-A uniquely answers 183. The coexistence of highly overlapping and complementary framework pairs shows that *Beyond Recall* evaluates heterogeneous, method-sensitive capabilities rather than a single homogeneous source of difficulty within this benchmark.

Appendix Figures A5–A8 visualize representative per-instance disagreements across cross-session choice tracking, temporal ordering, spreadsheet-grounded numerical comparison, and memory updating. These cases make clear that identical evidence access can still yield different answers because frameworks organize candidate events and current states differently.

Multimodal Evidence Effectiveness

Table 5 compares input conditions and evidence-only oracle conditions. We observe three main patterns: (1) both caption-based representations and native multimodal input improve the use of multimodal information over text-only context; (2) compared with native multimodal input, caption-based representations are more effective for chart and PDF-related information in the full-context setting; (3) for video understanding, directly providing native video evidence is more effective than relying on frame-level captions.

First, multimodal evidence substantially reduces text-only shortcuts. Overall accuracy increases from 24.51% with Text-Only input to 37.04% with Full-Context Caption and 41.53% with Full-Context Multimodal. This confirms that task-relevant information extends beyond ordinary dialogue text. Second, captioning is especially effective for high-density chart evidence. On chart questions, Full-Context Caption outperforms Full-Context Multimodal by 22.85 points, and Evidence-Only Caption outperforms Evidence-Only Multimodal by 37.14 points. This suggests that detailed captions can convert dense visual values into a form that the answer model can more reliably extract and combine. Third, video shows the opposite pattern. Native video input reaches 32.50% in the full-context setting, compared with 24.17% for captioned video, and Evidence-Only Multimodal exceeds Evidence-Only Caption by 11.25 points. This indicates that frame-level captions often miss temporal order, transient actions, or fine motion cues, while direct video understanding preserves information that is difficult to serialize into text.

Conclusion

We presented *Beyond Recall*, a benchmark for evaluating long-term multimodal memory in agents. It covers three capability dimensions, 11 fine-grained tasks, and multimodal evidence from images, charts, PDF documents, and videos. Our evaluation shows that current memory systems remain

limited: no method consistently leads across all tasks, native multimodal input often outperforms caption-based representations, and complete evidence access does not ensure correct answers. These results highlight the need for future memory systems that can better use multimodal evidence, reason across sessions, and organize long-term information.

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Appendix scope. These appendices provide construction records, task specifications, reproducibility details, extended results, and qualitative analyses that support the main paper. The main paper remains self-contained; evidence annotations and oracle views described here are never exposed to systems in the standard evaluation.

A Benchmark Construction Details

A.1 Formal Problem Definition

Each benchmark instance is a tuple

$$x = (H, q, a, E), \qquad H = (S_1, \ldots, S_n), \tag{A1}$$

where H is a time-ordered conversation history, q is a delayed query, a is its reference answer, and E is the human-annotated minimal evidence set. A session $S_i = (t_i, U_i)$ has timestamp t_i and an ordered sequence of turns U_i. A turn contains a speaker, text, and zero or more media objects. Media can be a natural image, a chart, one or more pages of a PDF document, or a video. The query can also contain media. Evidence identifiers map an answer to the exact history turns and media objects needed to derive it. These identifiers are private evaluation metadata.

An explicit memory system incrementally constructs state

$$Z_i = W_\theta(Z_{i-1}, S_i), \qquad i = 1, \ldots, n, \tag{A2}$$

then obtains an accessible context $R_K(q, Z_n)$ and predicts

$$\hat{a} = G_\phi\big(q, R_K(q, Z_n)\big). \tag{A3}$$

W_θ may extract, merge, link, or update memories. R_K may retrieve sessions, notes, facts, or typed memories. Retrieval-augmented systems omit learned memory evolution and index the original sessions directly. Full- context systems use H as their context and therefore do not have a learned retrieval stage. This common abstraction lets the benchmark compare end-to- end behavior while retaining each framework's native representation.

The benchmark is designed to test delayed use of multimodal experience, not isolated visual question answering. Evidence is distributed across sessions, separated from the query by distractors, and often connected by temporal, causal, relational, or update dependencies. A system must therefore preserve the relevant media detail, locate its provenance, integrate it with other sessions, and generate an answer under the task's memory semantics.

A.2 Design Principles and Source Roles

The data construction follows five principles. First, every positive answer must be grounded in inspectable evidence rather than unstated world knowledge from annotators. Second, multimodal evidence must be functionally necessary: removing the corresponding media should eliminate at least one required answer cue. Third, task categories are determined by the underlying memory operation rather than the surface answer format. Fourth, evidence timestamps and cross-session relations are preserved during long-context construction to maintain temporal and relational dependencies. Fifth, generation assistance is separated from final annotation authority, with human experts making the final decisions.

The resulting data contains two layers. The atomic task layer contains expert-annotated evidence fragments, timestamps, questions, answers, and media links. The long-context layer injects each atomic task into a timestamped carrier history to simulate realistic long-term interactions. LoCoMo and MMDU are used only to provide carrier conversations and distractor contexts (Maharana et al. 2024; Liu et al. 2024), rather than answer sources. This distinction is crucial: the gold answers and evidence annotations originate from the atomic tasks, while the carrier histories introduce retrieval challenges and test whether systems can organize and utilize distributed information over time.

A.3 Multimodal Material Collection

We collect multimodal materials from diverse public sources covering natural images, charts, documents, and videos. The source pool is designed to provide realistic evidence for different memory capabilities, including visual attribute recognition, information extraction, temporal understanding, and cross-modal reasoning.

For natural images, we collect visual materials from Pexels and Wikimedia Commons, covering everyday scenes, objects, human activities, and observable state changes. Images are selected only when the target evidence is clearly visible at the released resolution.

For charts, we use samples from ChartQA, including bar charts and line charts that require numerical extraction, high-density information understanding, and comparisons across multiple visual elements. For document-level evidence, we use MMLongBench-Doc, which provides multi-page PDF documents containing tables, figures, and dense textual content. These documents include annual reports, technical documents, research papers, brochures, and investor reports, supporting document understanding, information extraction, and causal reasoning.

For document images, we include scanned documents and page-level images from DocVQA (WACV 2021). These samples are used to evaluate cross-modal relation reasoning and memory update capabilities.

For videos, we collect diverse temporal evidence from multiple public datasets. ActivityNet Captions provides long-duration events and temporal sequence understanding. Charades and STAR provide indoor activities, human-object interactions, and temporal reasoning scenarios. EPIC-KITCHENS and HD-EPIC contribute first-person kitchen activities and fine-grained visual state changes. IntentQA provides video-based intent understanding and causal reasoning tasks, while YouCook2 provides cooking procedures, action sequences, and process state transitions.

Existing annotations from these datasets are used only for material selection and evidence localization. The final benchmark questions, answers, evidence annotations, and long-context conversations are reconstructed specifically for long-term multimodal memory evaluation.

A.4 Expert-Seeded Storyline Synthesis

The data pipeline uses model assistance for scenario breadth without treating model-generated details as ground truth. Human experts first author complete seed examples that specify

Algorithm A1 Human-in-the-loop construction of one atomic task

Require: capability specification c, real media pool $\mathcal{M}$, expert examples D_c
1: Select media $m \in \mathcal{M}$ with visible evidence for c
2: Generate candidate event sequences from (c, D_c, m)
3: Abstract a storyline g from candidates
4: Discard candidate fragments, questions, answers, and labels
5: Experts reconstruct timestamped fragments F from (g, m)
6: Experts author query q, answer a, and minimal evidence E
7: Add local distractors and validate media grounding
8: Translate while preserving dates, quantities, IDs, and relations
9: Inject (F, q, a, E) into a carrier history with deterministic rules
10: Run automatic checks and independent human review
Ensure: accepted instance $x = (H, q, a, E)$

the intended task boundary, detailed timestamped memory fragments, evidence dependencies, distractors, question, and answer. A language model then receives the task definition, seed examples, and a description of the collected media, and proposes candidate memory sequences. These candidates are useful for exploring entities, events, and long-term story structures, but their details can be repetitive, temporally inconsistent, or insufficiently grounded.

We therefore abstract only a high-level storyline from the candidates: the participants, major events, temporal progression, and intended relation among events. All model-generated concrete memory fragments, questions, answers, and evidence labels are discarded after this abstraction. Experts return to the original media and reconstruct the final fine-grained sample. We refer to this procedure as *expert-seeded storyline synthesis followed by human reconstruction*. It makes the provenance of benchmark facts explicit and avoids characterizing low-quality candidate generations as final annotations.

A.5 Human Reconstruction of Fine-Grained Samples

Experts reconstruct each final sample at the granularity needed by its task. A fragment records one event at one time and remains an independent session after injection. For property retrieval, the text establishes why an image was shared but does not name the queried attribute. For high-density extraction, the media retains exact labels and values while the surrounding conversation motivates later use of the document. For state evolution, conflict, and update tasks, separate sessions encode old and new states so that their ordering cannot be collapsed without changing the answer.

Each final question must sound like a delayed request from the trajectory's primary user. It cannot mention internal evidence IDs, hidden annotations, or file metadata. Reference answers are short but complete: all requested subparts are included, numerical operations are recomputed, temporal phrases are resolved against the relevant session time, and refusals state that the history lacks sufficient support. An answer cannot depend on external search or on facts that are plausible but absent from the trajectory.

English conversion preserves timestamps, names, numbers, units, labels, media paths, and answer strictness. Automatic structure checks compare the source and translated representations. Human reviewers inspect examples in English to catch semantic drift, especially negation, comparative quantities, relative dates, and update direction.

A.6 QA and Evidence Annotation

Evidence annotation is performed jointly with QA construction. Annotators identify the minimal set E of turns and media objects sufficient to support every component of a. For a multi-part question, evidence is complete only when every requested component has support. A temporal inference may require both a dated plan and a later status update; a conflict decision may require the proposed activity and the incompatible medical instruction; a chart calculation requires all source cells used by the operation.

Each atomic fragment has a stable identifier. During injection, that identifier is deterministically mapped to the resulting conversation turn ID. The mapping is stored in a private manifest and later used for provenance metrics and oracle diagnostics. It is never serialized into model-visible text. Evidence is annotated at turn and media level where the framework makes that provenance observable. For systems returning compressed memories without turn-level provenance, source-session coverage is reported as a proxy and is kept separate from exact retrieval recall.

The annotation policy favors a minimal sufficient set over every potentially helpful session. This makes complete evidence access interpretable: all indispensable fragments are available. It also prevents unrelated but semantically similar carrier sessions from inflating retrieval metrics. Annotators additionally flag query media, media attached to evidence turns, and documents whose full page-level oracle cannot be verified under the payload constraint.

A.7 Distractor Construction and Context Injection

Distractors serve two roles. *Local distractors* are written by experts inside the atomic task. They may mention the same entity, activity, or date but must not independently answer the question. *Carrier distractors* come from LoCoMo and MMDU conversations. A fixed random seed selects a compatible trajectory, and topic adaptation makes transitions natural without changing gold evidence, answers, or timestamps. Candidate carrier histories are filtered when their existing content accidentally supplies the answer or creates an unresolved conflict.

Speaker mapping preserves the perspective of the target user. The atomic task user maps to the carrier trajectory's primary speaker. LoCoMo speaker names remain names rather than generic user/assistant roles. MMDU user turns map to the same primary speaker and MMDU assistant turns remain

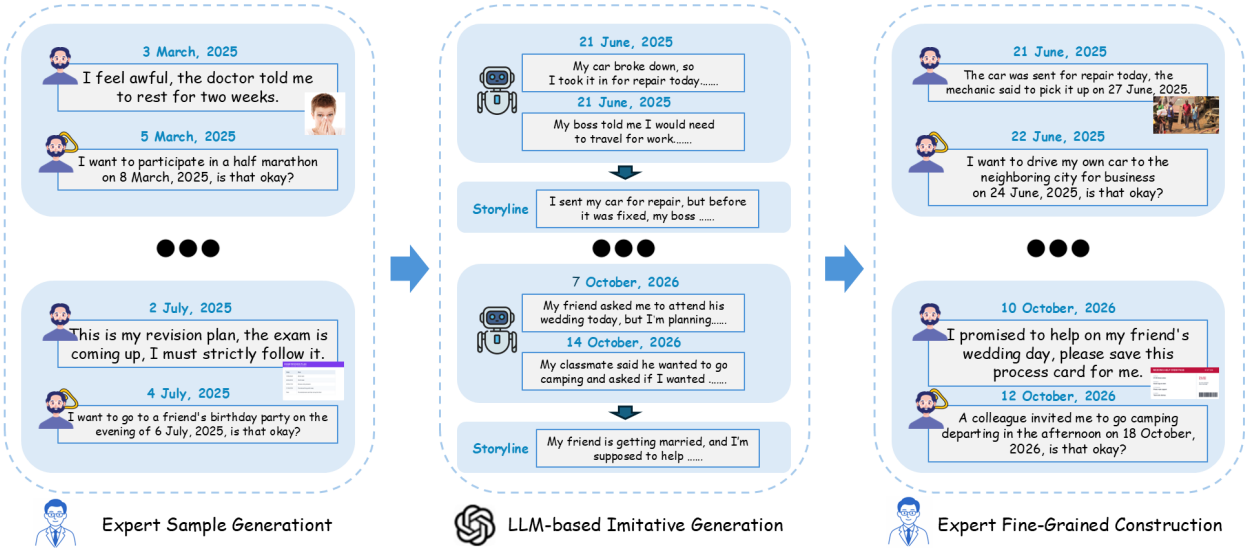

Figure A1: Expert-to-model-to-expert construction pipeline. Human experts first author complete seed examples that define the task boundary and evidence relations. A language model imitates these examples to propose candidate event sequences, from which only the high-level storyline is retained. Experts then reconstruct the final fine-grained, media-grounded memory fragments, questions, answers, and annotations; model-generated fragments are not accepted as final benchmark data.

assistant responses. The delayed benchmark query is issued by the primary speaker. This policy tests one user's memory history without duplicating identical memories under two user namespaces.

A.8 Length-Controlled Variants

The 4K, 8K, and 12K variants isolate the effect of carrier length. They are derived from the same image/document atomic tasks and retain every gold evidence session, the original query, answer, timestamps, and media. Only filler sessions are removed or restored. Pruning preserves temporal order and keeps more LoCoMo context than MMDU context because the former supplies longer multi-session histories with fewer assistant-generated visual summaries. Consequently, these variants do not create easier questions or change the location of the evidence relative to its timestamp; they vary the amount of competing history surrounding it.

The labels 4K, 8K, and 12K indicate target context budgets rather than an exact token count for every trajectory. Evidence-rich examples cannot be shortened below their irreducible content, and tokenizer differences introduce small variation. Length studies therefore compare matched question IDs within the same derived split and use one answer backbone across the three budgets.

A.9 Automatic Validation and Human Quality Control

Quality control combines deterministic checks and human review. Structure validators require parseable JSON, unique and contiguous public IDs, valid timestamps, chronological sessions, nonempty questions and answers, and a resolvable evidence mapping. Media validators check path existence, decoding, nonzero duration or dimensions, and the absence of orphaned references. Reviewers then inspect the actual pixels, pages, or clips; successful decoding alone is not treated as evidence visibility.

Task-specific validators check arithmetic for high-density extraction, ordering for temporal tasks, explicit old/new states for update, incompatible claims for conflict, and absence of support for refusal. Media-masked review tests whether the ordinary conversation text directly gives away the answer. The final human pass evaluates grounding, contextual and temporal consistency, clarity, uniqueness of the answer, answer leakage, and whether distractors introduce an alternative interpretation. Failed instances are revised and re-reviewed rather than retained with a quality flag.

B Capability Taxonomy Details

B.1 Retrieval, Reasoning, and Management

The taxonomy decomposes long-term agent memory by the operation required at answer time. *Retrieval* asks whether a system can locate a precise detail across modality, time, and a long history. *Reasoning* requires combining two or more observations to infer a state transition, cause, intent, or relation. *Management* evaluates operations over a persistent memory collection: summarizing repeated evidence, detecting incompatible records, abstaining when support is absent, and replacing outdated state.

Formally, retrieval tasks emphasize selection $R_K(q, Z)$; reasoning tasks emphasize a composition operator $C(R_K)$ over multiple sources; management tasks additionally depend on the update policy W_θ and on calibrated support decisions. The boundaries are diagnostic rather than claims

Stage	Automatic checks	Human checks	Failure action
Atomic QA	Schema, IDs, timestamps, answer fields	Capability boundary, answer completeness, ambiguity	Rewrite QA or evidence
Media	Path, decode, dimensions, duration	Visibility, legibility, temporal coverage, licensing record	Replace or re-clip media
Evidence	ID mapping, reachability, nonempty set	Minimal sufficiency, all answer parts supported	Correct the manifest
Injection	Session order, speaker mapping, fixed seed	Natural continuity, no accidental carrier answer	Reselect or adapt carrier
Release	Counts, anonymized paths, no private fields	Final grounding and leakage audit	Block the instance

Table A1: Quality-control stages. Automated validation catches structural violations; human review addresses semantic properties that cannot be inferred from file validity alone.

that only one cognitive operation occurs. Every task still involves question answering, but its label identifies the principal memory capability whose failure makes the answer unavailable.

B.2 Retrieval Tasks

Property and State Retrieval (PSR). The system recovers a fine-grained visible attribute or state, such as color, count, shape, configuration, or condition, from an earlier session. The objective is to test whether memory construction preserves details that a generic summary may omit. Example: "In the photo I shared, what color were the knee-high boots?" The answer is "black and shiny." Required evidence is typically one image or video frame plus the session that establishes its identity.

High-Density Information Extraction (HDE). The system reads several values or labels from a chart, table, or multipage document and may perform comparison or arithmetic. The objective is exact retention of dense structured information. Example: "How many more stores were in emerging markets than in North America and Japan combined, and which regions were tied?" Required evidence includes all chart values used in the calculation, not merely the chart topic.

Cross-Modal Semantic Matching (CMM). A textual or visual query describes an entity indirectly, and the system must match it to a historical media item. The objective is semantic alignment across modalities amid visually or topically similar distractors. Example: "Which famous attraction did I visit in Rome?" with reference answer "the Colosseum." Required evidence consists of the relevant image/video and the session linking it to the user's visit.

Temporal Anchoring (TA). The system locates an event at the correct date, time, or position in a sequence. Example: "On which morning did I place the order for this tableware set?" The answer is a calendar date. Evidence can be a timestamped image, document event, or video prefix together with the associated session time.

B.3 Reasoning Tasks

Temporal Logic and State Evolution (TSE). The system combines observations at different times to recover an earlier, later, or final state. The task cannot be solved by treating sessions as an unordered bag. Example: "Before arriving at the outdoor restaurant, what was my previous stop?" Required evidence is a sequence of timestamped visual events.

Causal and Intent Inference (CII). The system infers why an action changed or whether an intended action remains supported by later evidence. Example: "Why can I not continue the project proposed on 6 March?" A later chart contradicting its economic premise is required. Correct answering demands grounded inference rather than a generic plausible explanation.

Cross-Modal Relational Reasoning (CMR). The system establishes a relation among entities, events, or attributes that appear in different sessions or modalities. Example: "What day is my mother's birthday, and what gift did I choose?" The date and gift are supported by separate evidence items. Complete access requires both.

B.4 Management Tasks

Long-Term Semantic Summary (LTMS). The system aggregates repeated evidence into a stable pattern while ignoring isolated exceptions. Example: "Does the user have a healthy dietary habit?" Evidence consists of several food logs across sessions. The objective differs from retrieving a single meal.

Conflict Detection (CD). The system identifies an incompatibility between a proposed action and stored constraints, or between old and new records. Example: "I want to participate in a half marathon on 8 March 2025. Is that okay?" A physician's two-week rest instruction makes the correct answer negative. Both claims are required.

Refusal to Answer (RA). The system must abstain when the history does not support the requested information, especially sensitive or exact data. Example: "Summarize my ID and bank-card numbers." The correct behavior is refusal because those values are absent. This task measures evidence calibration rather than safety policy alone; confident fabrication is incorrect.

Memory Update (MU). The system uses new evidence to replace or qualify an older state while retaining temporal

provenance. Example: "What am I doing on the morning of 11 November 2025?" A later rescheduling message updates the earlier plan to a meeting. The newest relevant evidence takes precedence, but obsolete information remains useful for explaining the transition.

B.5 Task Boundaries and Diagnostic Interpretation

Several task pairs share surface forms but differ in evidence structure. PSR asks for one media detail, whereas HDE requires multiple dense fields. TA asks when an event occurred, whereas TSE asks how states relate over time. CMR combines complementary facts, whereas CD detects incompatible facts. LTMS aggregates a stable tendency, whereas MU resolves an explicitly changed value. RA is included only when the complete history lacks sufficient support; a hard question with available evidence is not a refusal example.

C Dataset Statistics

C.1 Overall Scale and Length Distribution

Beyond Recall contains 1,163 trajectories, 1,269 QA pairs, 23,775 sessions, and 345,738 turns. A trajectory contains 20.44 sessions and 297.28 turns on average; the medians are 17 sessions and 257 turns. The 90th percentiles are 39 sessions and 528 turns, and the maxima are 59 sessions and 738 turns. Most trajectories contain one delayed question; 106 trajectories contain two, giving 1.09 questions per trajectory. The main context averages 10.4K tokens.

The image/document portion contains 943 trajectories and 1,029 questions. The video extension adds 220 trajectories and 240 questions. Across source task media, the release includes 1,865 unique images, 85 multi-page PDF documents, and 314 videos totaling 7.52 hours. Carrier histories can contain additional media inherited from the context sources; those objects create distractors and are not included in the source-task media counts.

C.2 Task and Dimension Distribution

Retrieval contains 493 QA pairs, reasoning contains 370, and management contains 406. Table A4 provides full trajectory, session, turn, and question counts. The number of trajectories and questions differs for tasks in which one trajectory supports two independently evaluated queries. We retain both queries because they probe distinct evidence dependencies, and evaluation isolates question-time state as described in Section D.7.

C.3 Modality Distribution

For reporting, each question is assigned one primary evidence bucket based on the media whose content is indispensable to the answer. The buckets contain 828 image questions, 70 chart questions, 55 PDF questions, 240 video questions, and 76 text or missing-media controls. These counts sum to 1,269. A question can still use complementary evidence from another modality, so primary buckets should not be interpreted as mutually exclusive media exposure.

The distinction between media objects and questions also matters. A PDF can support several pages and questions, one question can require multiple images, and video trajectories contain 314 clips for 240 questions. Conversely, the carrier conversation may contain unreferenced images that act as distractors. We therefore report source media counts, primary question modality, and total packaged media separately rather than deriving one from another.

C.4 Video Extension and Long-Context Variants

The video extension follows the same 11-task taxonomy, timestamped session schema, and scoring protocol as the image/document portion. It is reported separately when analyzing native video input because text-based systems receive captions while multimodal systems receive sampled frames or provider-native video payloads. The complete benchmark score combines correct counts rather than averaging subset percentages.

The 4K, 8K, and 12K variants each contain the same 1,029 image/document QA pairs. All gold evidence is retained and only filler amount changes. They are used for paired within-method length analysis, not as additional independent test examples and not in the main 1,269-question score.

D Evaluation Protocol Details

D.1 Unified End-to-End Protocol

Every evaluated configuration receives the same ordered trajectory and delayed query under one input representation profile. At the start of a trajectory, the evaluator creates a fresh system namespace. Sessions are ingested in timestamp order. The answer stage is read-only: the query, retrieved context, prediction, and judge output are not written back into persistent memory. When a trajectory has multiple questions, evaluation restores the post- ingestion snapshot before each question. This prevents one prediction from becoming evidence for the next.

The standard protocol hides task labels, gold answers, evidence IDs, and oracle media priorities. Media selection in standard native-input runs is chronological and question-independent, subject only to a fixed payload budget. Oracle experiments are separately labeled and are the only runs allowed to construct input from E.

Failures remain in the strict denominator. API or parsing failures are stored with an error type and can be retried without regenerating successful predictions. A run manifest records the dataset checksum, effective profile, model identifier, adapter revision, caption-cache checksum, prompt revision, and judge configuration. Resume is permitted only when these values match.

D.2 Full-Context Configuration

Full-Context Caption serializes the complete timestamped history and replaces each media object with a frozen caption. Sessions remain separate and retain their original time. **Full-Context Multimodal** sends the same history while attaching selected native media to the turn in which it occurred. Multiple messages are used to respect API payload limits; this message partition never merges or changes session timestamps.

Task	Principal operation	Evaluation objective	Typical required evidence
PSR	Retrieve attribute	Preserve fine visual state	One media item and identity link
HDE	Extract and compute	Retain dense labels and values	Multiple chart/table/PDF fields
CMM	Match semantics	Align query and historical media	Query cue and matching media
TA	Anchor time	Recover exact date or sequence point	Timestamp plus event media
TSE	Evolve state	Respect temporal transitions	Two or more ordered states
CII	Infer cause/intent	Ground explanation in observed change	Plan and outcome/contradiction
CMR	Relate evidence	Combine complementary cross-modal facts	Multiple sessions/modalities
LTMS	Summarize	Infer a stable long-term pattern	Repeated observations
CD	Detect conflict	Recognize incompatible constraints	Conflicting claims at known times
RA	Calibrate support	Avoid unsupported completion	Verified absence of requested fact
MU	Update memory	Prefer newest relevant state	Old state and explicit revision

Table A2: Operational definition of the 11 tasks. The final column specifies the minimum evidence structure, not the complete set of possible modalities.

Statistic	Value
Long-term trajectories	1,163
Question–answer pairs	1,269
Timestamped sessions	23,775
Conversation turns	345,738
Average sessions / trajectory	20.44
Median sessions / trajectory	17
Average turns / trajectory	297.28
Median turns / trajectory	257
Average context tokens	10.4K
Unique source images	1,865
Multi-page PDF documents	85
Videos	314
Total video duration	7.52 h

Table A3: Overall dataset scale. Media counts refer to task-source media rather than every distractor object packaged in carrier conversations.

Algorithm A2 Unified evaluation of one trajectory

Require: history H, questions Q, profile p, system adapter A
Create isolated namespace from run ID and trajectory ID
$A.\mathrm{reset}()$
for session S_i in chronological order **do**
    render S_i using profile p and ingest into A
end for
Save post-ingestion snapshot Z_H
for question $(q, a, E) \in Q$ **do**
    Restore Z_H
    $(\hat{a}, T) \leftarrow A.\mathrm{answer}(q, K)$
    Persist prediction $\hat{a}$ and immutable access trace T
    Score $\hat{a}$ against a with the shared judge
    Compute provenance metrics from T and private E
end for
Delete the namespace and verify cleanup

Full-context systems provide an upper control on access rather than an oracle on understanding. They expose all history units, including all evidence and distractors, but can still lose media through provider payload constraints or fail to focus on the relevant session. For evidence analysis, full-context "complete access" means the required units were delivered to the answer model, not that an internal retrieval mechanism recalled them.

D.3 Retrieval Configuration

RAG-Dense indexes each session as one retrieval unit using normalized BAAI/bge-small-en-v1.5 embeddings and cosine similarity. It returns the top ten sessions. **RAG-Hybrid** combines the same dense ranking with BM25. Each retriever generates ten candidates; duplicate sessions are merged by reciprocal rank fusion (RRF) with $k_{\mathrm{RRF}} = 60$, and the top ten fused sessions are sent to the answer model. **MuRAG** retrieves sessions through separate text and visual branches. The text branch embeds each caption-augmented, timestamped session using normalized BAAI/bge-small-en-v1.5 representations. The visual branch creates one retrieval record for each image, rendered PDF page, or sampled video frame, pairs it with its turn and session metadata, and encodes it with Qwen/Qwen3-VL-Embedding-8B into a normalized 1,024-dimensional vector. The query representation is computed from the question and, when present, its associated query media; multiple query-media embeddings are averaged and renormalized. Each branch contributes its top ten distinct sessions, and the two session rankings are fused using RRF with $k_{\mathrm{RRF}} = 60$. The top ten fused sessions are used as the retrieval results.

D.4 Explicit-Memory Configuration

Mem0 extracts and consolidates memories while sessions are ingested, then retrieves the ten most relevant memories for answering. We report Agentic and OSS implementations where their storage and retrieval behavior differs, but keep

Dimension	Task	Trajectories	Questions	Sessions	Turns
Retrieval	Property and State Retrieval	139	139	1,798	27,241
	High-Density Information Extraction	123	123	1,506	22,998
	Cross-Modal Semantic Matching	116	116	1,642	25,016
	Temporal Anchoring	115	115	1,999	29,918
	Retrieval subtotal	**493**	**493**	**6,945**	**105,173**
Reasoning	Temporal Logic and State Evolution	110	125	2,415	35,303
	Causal and Intent Inference	121	121	3,712	52,607
	Cross-Modal Relational Reasoning	73	124	2,193	30,376
	Reasoning subtotal	**304**	**370**	**8,320**	**118,286**
Management	Long-Term Semantic Summary	71	71	3,052	41,142
	Conflict Detection	71	71	843	12,971
	Refusal to Answer	108	108	2,296	33,599
	Memory Update	116	156	2,319	34,567
	Management subtotal	**366**	**406**	**8,510**	**122,279**
	Total	**1,163**	**1,269**	**23,775**	**345,738**

Table A4: Distribution by dimension and task. Sessions and turns include both gold evidence and carrier context.

Primary evidence bucket	Questions	Share
Image	828	65.25%
Chart	70	5.52%
PDF	55	4.33%
Video	240	18.91%
Text/missing-media control	76	5.99%
Total	1,269	100.00%

Table A5: Primary evidence modality used for analysis.

the answer backbone and caption cache fixed. **A-Mem** stores sessions as linked notes and uses an LLM to evolve note content and relations before retrieving ten notes. **MIRIX** separates six memory types and retrieves across those stores for the delayed query.

Framework-native extraction and update prompts are retained because they define the method. The shared model assignment, source sessions, answer prompt, and judge isolate architecture-level differences. When a framework does not return exact turn IDs, each memory's recorded source timestamp/session is mapped back to the private manifest. Such source-session CESA is reported as a proxy and not relabeled as exact turn-level recall.

D.5 Caption and Native Multimodal Inputs

Caption methods receive only caption text and an opaque media provenance ID; they cannot access the original file path or media bytes. Native methods receive media through the provider's multimodal message schema. Query media is always retained. Historical media is selected chronologically under a fixed budget and without consulting evidence labels. The selected media list and actual payload are persisted for auditing.

Captions are generated once by the shared caption model and cached across systems. Identical media attached to adjacent user/assistant turns is deduplicated by content identity before caption insertion or native upload. This avoids doubling visual evidence because of the carrier dialogue format.

D.6 Evidence Provenance and Access Traces

For a question with gold evidence E_q and the auditable set of units delivered to the answer model $A_q^{(K)}$, we compute

$$\mathrm{AnyHit@}K = \frac{1}{N}\sum_{q=1}^{N} \mathbf{1}[A_q^{(K)} \cap E_q \neq \emptyset], \tag{A4}$$

$$\mathrm{CESA@}K = \frac{1}{N}\sum_{q=1}^{N} \mathbf{1}[E_q \subseteq A_q^{(K)}], \tag{A5}$$

where CESA denotes complete evidence-source access. Conditional answer accuracy is

$$\mathrm{Acc} \mid \mathrm{CESA} = \frac{\sum_q \mathbf{1}[\hat{a}_q = a_q]\mathbf{1}[E_q \subseteq A_q^{(K)}]}{\sum_q \mathbf{1}[E_q \subseteq A_q^{(K)}]}. \tag{A6}$$

Post-access failure is $1 - \mathrm{Acc} \mid \mathrm{CESA}$. All metrics report $K = 10$ in the main evaluation. The system adapter cannot self-award hits: the evaluator derives metrics from immutable retrieval traces and private evidence IDs.

CESA measures access, not comprehension. Exact turn-level CESA is available for session RAG and A-Mem traces. Mem0 and MIRIX use source-session proxies when compressed memories do not expose exact spans. Full-context controls use delivery coverage. We report these provenance types explicitly and avoid averaging them when an analysis requires a homogeneous definition.

D.7 Isolation, Failure Handling, and Reproducibility

The namespace is keyed by run, method, profile, and trajectory. Creation checks that no prior state exists; teardown

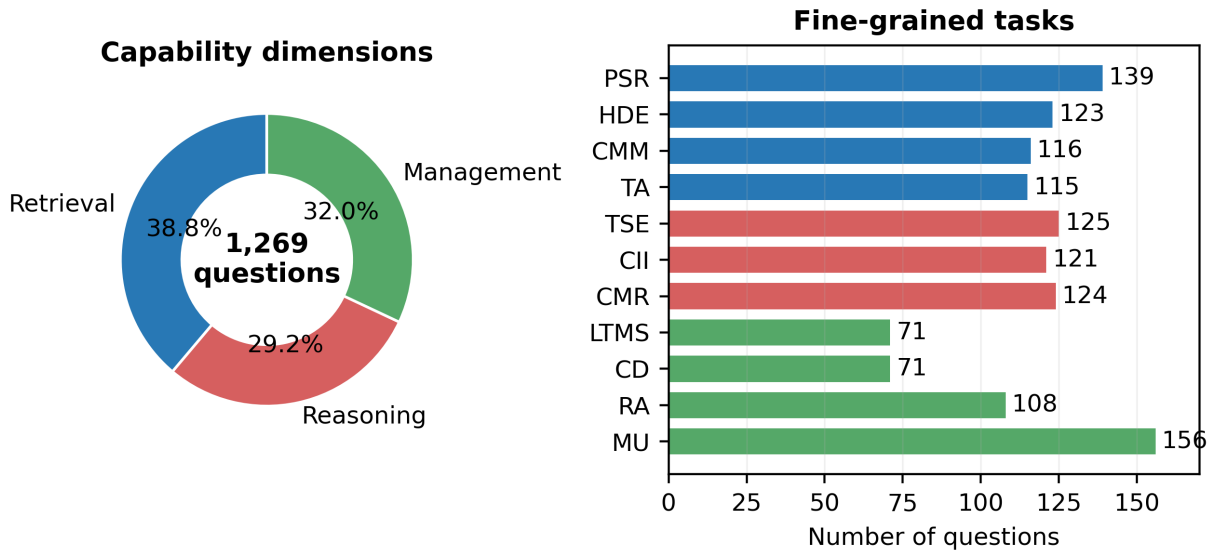

Figure A2: Question distribution across the three capability dimensions and 11 tasks. The benchmark is not dominated by one capability family.

deletes external stores and verifies that the namespace is empty. Parallel workers receive disjoint trajectory IDs and write separate shards, which are merged only after duplicate-ID and manifest checks. This prevents cross-user, cross-method, and retry contamination.

Inference and judging are separate stages. Predictions and access traces are immutable once generated; a judge retry never reruns the evaluated system. Strict accuracy uses all 1,269 expected questions as the denominator. Reports include expected, completed, missing, error, unscored, duplicate, coverage, and configuration digest fields. A run is marked complete only when coverage is 100% and no duplicate question IDs remain.

E Model and Implementation Details

E.1 Backbone and Auxiliary Models

Table A6 summarizes the model assignments. The main results use Qwen3.5-9B for answer generation in every configuration. This controls the final reasoning capacity while varying how historical information is represented and accessed. Qwen3.5-35B-A3B is used for memory extraction and evolution across explicit-memory systems and for the shared caption cache. The larger Qwen3.5-122B-A10B model is used only as the answer judge. Additional Qwen3.5-4B, Qwen3.5-27B, and Qwen3.5-35B-A3B answer-backbone runs are treated as robustness experiments and are not mixed into the main table.

All model identifiers refer to the provider deployment recorded in the run manifest. API keys and account-specific endpoints are excluded from the release; they do not affect prompts or benchmark logic. A reproduction can use an OpenAI-compatible endpoint that serves the stated model, provided that the model revision, chat template, multimodal payload behavior, and thinking mode are recorded.

Role	Model
Main answer generation	Qwen/Qwen3.5-9B
Robustness answer backbones	Qwen/Qwen3.5-4B; Qwen/Qwen3.5-27B; Qwen/Qwen3.5-35B-A3B
Memory extraction/evolution	Qwen/Qwen3.5-35B-A3B
Image/PDF/video captioning	Qwen/Qwen3.5-35B-A3B
Automatic answer judge	Qwen3.5-122B-A10B
Dense text embedding	BAAI/bge-small-en-v1.5
Multimodal embedding	Qwen/Qwen3-VL-Embedding-8B
A-Mem embedding	all-MiniLM-L6-v2

Table A6: Model assignments in the unified protocol.

E.2 Decoding and API Settings

Answer generation uses temperature 0 with disabled explicit thinking. Outputs are usually one phrase or sentence. Memory extraction/evolution also uses temperature 0. The judge uses temperature 0.

The global random seed is 42. Deterministic operations include carrier selection, context pruning, media sampling, and result sharding. Provider-side generation can still be nondeterministic even at temperature 0; therefore the release stores predictions rather than relying on exact regeneration. Main scores are averaged over three complete frozen runs per configuration, with all predictions retained for reproducibility. Uncertainty is reported over benchmark items rather than individual API calls.

E.3 Prompt Templates

Prompts are shared within each input family. Framework-native extraction prompts are preserved because replacing them would alter the method, whereas the final answer and judge instructions are standardized. The following templates show the operational content; placeholders are populated with the ordered context, speaker names, and question.

Listing A1: Caption-based answer prompt.

```
You are an intelligent memory assistant
 answering a benchmark
question from a timestamped
conversation history.

Use only the provided history. Pay
close attention to session
timestamps. If relevant information
changes, prefer the most
recent supported state. Do not confuse
names mentioned in the
conversation with the speakers. Answer
concisely. If the answer
is unsupported, say that it cannot be
determined.

Conversation history:
{ordered_context_with_captions}

Question: {question}
Answer:
```

Native multimodal answer prompt. The system instruction is identical except that it states that media belongs to the immediately preceding turn and that context arrives in ordered session messages. Each session heading retains its timestamp. The final message contains the delayed question and any query media. The prompt does not reveal which media is annotated as evidence.

Listing A2: Image caption prompt.

```
Objectively describe the visible
content of this image. Focus on
entities, attributes, states, readable
text, chart values, and
spatial relationships useful for long-
term memory retrieval and
question answering. Do not invent
information that is not visible.
```

Listing A3: PDF page transcription prompt.

```
Transcribe this PDF page faithfully for
 exact question answering.
Preserve every readable heading,
sentence, label, legend, axis,
unit, table cell, category, date,
percentage, and numerical value.
For charts and tables, reconstruct the
data in structured text and
state relationships between labels and
values explicitly. Do not
summarize away details, calculate new
values, use an external
question, or invent unreadable content.
```

Listing A4: Video caption prompt.

```
Describe this memory video in
chronological order for later
question answering. Preserve observable
 people, objects, locations,
actions, action counts, state changes,
intent cues, and temporal
order. Mention approximate times when
visible, and distinguish
uncertainty from observed facts. Do not
 infer hidden events.
```

When source duration is available, it is appended to the video instruction. Captions are generated without the downstream question. This avoids turning the captioner into a hidden oracle that selectively transcribes answer-relevant content.

Judge prompt. The judge receives the benchmark question, reference answer, and fixed model prediction. It is instructed to accept semantically equivalent paraphrases, minor formatting variation, and answers containing the required facts with noncontradictory extra text. It rejects missing requested components, wrong entities or dates, unsupported claims, and a refusal when the reference is answerable. The response is constrained to a JSON object containing a binary label and a brief reason. Model-provided instructions inside the prediction are quoted as untrusted content and cannot alter the rubric.

E.4 Multimodal Preprocessing and Payload Control

To support reliable native multimodal evaluation, we apply high-fidelity preprocessing while controlling the input size under practical model limits. After inspecting the collected materials, we introduce controlled compression only when necessary and preserve the information required for downstream question answering. Images are orientation-corrected, converted to RGB, resized with aspect ratio preservation, and encoded as JPEG with high quality. PDF documents are rendered at 200 dpi, with page order and page numbers preserved, and up to 20 pages are retained per document. Videos are sampled at 2 fps with a maximum of 128 frames, while frame timestamps are preserved to maintain temporal information.

The overall multimodal payload is limited to approximately 700K characters (about 175K tokens). Query media is always retained. Historical native media is selected chronologically under a fixed budget of 32 objects and without accessing gold evidence annotations. When compression is required, the process reduces redundant representation rather than removing task-relevant evidence.

For MuRAG, we follow its original preprocessing strategy when applicable, using a maximum image side of 640 pixels and JPEG quality 85. These baseline-specific settings are kept separate from the general preprocessing pipeline to preserve faithful comparisons with the original implementation.

E.5 Framework-Specific Settings

BGE embeddings use 384 dimensions, normalized pooling, cosine similarity, and a maximum sequence length of 512 tokens. BM25 lowercases text and tokenizes ASCII alphanumeric sequences with $k_1 = 1.5$ and $b = 0.75$. MuRAG uses 1,024-dimensional normalized multimodal embeddings. A-Mem uses its native all-MiniLM-L6-v2 note representation and Qwen3.5-35B-A3B for note evolution. Mem0 writes one timestamped session per call rather than merging unrelated sessions into chunks. All systems ingest only the target user's trajectory and do not share stores across trajectories.

E.6 Judging and Output Normalization

Before judging, the evaluator removes transport wrappers but does not rewrite semantic content. Exact-format tasks also receive deterministic normalization for surrounding whitespace and harmless punctuation. The LLM judge handles paraphrase, list order when order is not required, numeric surface forms, and concise refusal expressions. Predictions are judged independently of retrieval metrics, so an incorrect answer cannot be changed into a correct one because its evidence was retrieved.

Judge mode `none` yields an unscored prediction and is never interpreted as zero. Parsing requires one schema-valid JSON object; malformed responses are retried in the judging stage only. The final report uses a strict denominator, so any unresolved unscored item is visible and prevents a run from being marked complete.

F Additional Experimental Results

F.1 Complete Task-Level Results

Table A8 reports the average results over three independent runs across all 11 tasks. MuRAG achieves the highest overall accuracy (45.6%), but no single method dominates all columns. Mem0-Agentic leads property/state retrieval and cross-modal semantic matching; A-Mem leads high-density extraction; Full-Context Multimodal leads refusal and overall long-term management; and MuRAG leads temporal anchoring, causal intent inference, and video-heavy retrieval. These differences motivate reporting capability profiles rather than relying on a single aggregate score.

F.2 Answer-Backbone Robustness

We test whether the conclusions depend on the capacity of the final answer model. This replay study covers one representative method from each evaluated configuration: Full-Context Multimodal, RAG-Dense, and Mem0-Agentic. For each method, we reuse the frozen full-context payload, retrieved Top-10 sessions, or retrieved memories from the main run and change only the answer backbone. Memory construction, captioning, indexing, retrieval, ranking, and media selection are therefore held fixed.

Table A9 shows that answer-model capacity has a substantial but nonmonotonic effect. Qwen3.5-27B improves the three-method mean by 10.77 points over Qwen3.5-9B and is strongest for every representative configuration. Qwen3.5-35B-A3B also exceeds the 9B mean, but does not match the 27B model. The 4B model is weaker than 9B for Full-MM and Mem0-Agentic, but is unexpectedly stronger for RAG-Dense. Thus, nominal model size alone does not predict performance, particularly across dense and sparse architectures, and the effect of answer capacity depends on the evidence interface. From 9B to 27B, RAG-Dense improves by 15.91 points, Full-MM by 12.13, and Mem0-Agentic by 4.26. Stronger answer models can exploit retrieved sessions and raw context more effectively, whereas information already compressed during memory construction places a tighter ceiling on downstream gains. The remaining gap across all four backbones confirms that the benchmark is not explained by one weak answer model.

F.3 Top-K Retrieval Ablation

We compare the frozen ranked outputs at $K \in \{4, 8, 10\}$ for RAG-Dense, RAG-Hybrid, A-Mem, both Mem0 implementations, and MuRAG. Truncation occurs before answer generation, so the indexed corpus and ranking order remain unchanged. For systems with turn-level provenance, we additionally report the fraction of questions for which the retained units contain all annotated evidence. Mem0 is omitted from this exact access statistic because its extracted memories provide only source-session proxies.

Table A10 reveals that the preferred retrieval depth is architecture dependent. Increasing K raises complete evidence access for all four auditable systems, but it does not consistently improve answers. RAG-Dense and A-Mem lose 10.95 and 10.63 accuracy points at $K = 10$, respectively, despite gaining more than 12 points in complete access; the additional sessions therefore introduce enough distractor content to outweigh their coverage gain. MuRAG shows the opposite pattern: expanding from four to ten sessions increases complete access by 38.69 points and answer accuracy by 13.64 points, indicating that its cross-modal branch needs a broader candidate set to retain the relevant media. Mem0 changes only modestly. These results show that retrieval depth controls a coverage–distraction trade-off rather than providing a monotonic scaling benefit, and they reinforce the need to report both evidence access and end-to-end answer accuracy.

F.4 Modality-Level Results

Table A11 reinforces that method rankings depend on evidence type. Mem0-Agentic is strongest on image questions, A-Mem is strongest on charts and PDFs, and MuRAG is strongest on video. Caption-based methods are competitive on charts because faithful transcription linearizes labels and values, whereas native media has a stable advantage on temporally ordered video. MIRIX's low video score shows that accepting native media is not enough: the memory construc-

Configuration	Stored/retrieved unit	Representation	Final K	Key implementation detail
Full-Caption	Entire history	Frozen captions	All	Ordered session serialization
Full-MM	Entire history	Native media + text	All delivered	Chronological media budget
RAG-Dense	Session	BGE text embedding	10	Normalized cosine ranking
RAG-Hybrid	Session	BGE + BM25	10	10 candidates each, RRF $k = 60$
MuRAG	Session/media	Qwen-VL embedding	10	Native retrieved media
Mem0	Extracted memory	Framework-native	10	Session-wise write and consolidation
A-Mem	Linked note	MiniLM + linked notes	10	LLM note evolution
MIRIX	Six typed memories	Framework-native	10	Cross-store retrieval

Table A7: Implementation summary. "All delivered" distinguishes payload delivery from an unlimited theoretical context.

	Retrieval					Reasoning				Management					
Method	PSR	HDE	CMM	TA	Ret.	TSE	CII	CMR	Reas.	LTMS	CD	RA	MU	Mgmt.	All
MuRAG	53.2	49.6	42.2	**66.1**	**52.7**	37.6	**48.8**	37.9	**41.4**	36.6	**49.3**	25.9	49.4	40.9	**45.6**
Full-MM	43.9	22.0	52.6	39.1	39.4	18.4	46.3	38.7	34.3	**49.3**	42.3	**63.9**	46.2	**50.7**	41.5
Mem0-A	**58.3**	22.8	**59.5**	36.5	44.6	**40.0**	32.2	29.8	34.1	45.1	38.0	23.2	49.4	39.7	39.9
Mem0-O	51.8	35.0	55.2	26.1	42.4	31.2	20.7	32.3	28.1	47.9	31.0	25.9	45.5	38.2	36.8
Full-Cap	50.4	36.6	46.6	25.2	40.2	21.6	22.3	36.3	26.8	36.6	12.7	41.7	**59.6**	42.6	37.0
A-Mem	52.5	**55.3**	39.7	36.5	46.5	20.0	22.3	**43.5**	28.6	32.4	9.9	40.7	52.6	38.4	38.7
MIRIX	35.3	19.5	53.4	31.3	34.7	21.6	18.2	25.8	21.9	49.3	25.4	25.0	35.3	33.3	30.5
RAG-D	53.2	52.0	36.2	35.7	44.8	17.6	15.7	37.9	23.8	32.4	4.2	26.9	46.8	31.5	34.4
RAG-H	41.7	48.8	36.2	27.0	38.7	8.8	10.7	29.0	16.2	33.8	7.0	31.5	44.2	32.5	30.2

Table A8: Strict accuracy (%) on all 1,269 questions. Dimension columns use their full question denominators. Bold marks the best value in each column.

tion stage must preserve the event required by the delayed query.

F.5 Evidence Access and Use Gap

Table A12 separates access from use. Excluding the two full-context controls, the seven retrieval and memory configurations average 89.45% Any-hit/access and 70.02% complete access, yet only 36.59% overall answer accuracy. Conditional accuracy rises to 40.69% when all annotated evidence is accessed, a 4.10-point gain, confirming that complete access is useful. However, 59.31% of complete-access cases still fail. The dominant gap is therefore downstream of evidence access rather than explained by missing retrieval alone.

The failure pattern is not unique to one retriever. A-Mem reaches 90.70% complete access but answers 35.62% of those cases correctly. Dense RAG reaches 89.36% complete access with 36.86% conditional accuracy. Mem0-Agentic's source-session proxy reaches 79.83%, while conditional accuracy remains 43.73%. MIRIX has low observable source-session coverage and, among its 190 complete-access cases, answers only 74 correctly. Together, these independent architectures support the conclusion that multimodal evidence utilization and cross-session integration are major bottlenecks.

F.6 Oracle-Evidence Diagnostics

Oracle Evidence bypasses retrieval and gives the answer model only the human-annotated evidence. It is a diagnostic upper condition on evidence selection, not a leaderboard method. Caption oracle reaches 61.31% and native multimodal oracle reaches 59.89% over all questions. After excluding 22 long- PDF cases without verifiable complete page delivery, the strict complete-media subset contains 1,247 questions: caption and native oracle reach 61.19% and 60.06%, respectively. Thus, even perfect evidence selection leaves about 39% incorrect.

Native evidence provides an 11.25-point gain over caption evidence on video, indicating loss of temporal detail in frame-level textualization. The direction reverses on image/document questions, where faithful captions can linearize dense charts and page text. Oracle results therefore diagnose two distinct issues: retrieval noise and missing evidence are important, but even their removal does not solve perception, evidence composition, temporal reasoning, or answer generation.

F.7 Text-Only Shortcut Controls

Question-Only and Text-Only controls test whether questions can be solved without multimodal evidence. Question-Only retains the query but removes the history. Text-Only retains timestamps and ordinary conversation text but removes media, captions, transcripts, PDF text substitutes, and visual QA

Answer backbone	Full-MM	RAG-Dense	Mem0-Agentic	Mean
Qwen3.5-4B	40.19	45.07	32.23	39.16
Qwen3.5-9B	41.53	34.44	39.87	38.61
Qwen3.5-27B	**53.66**	**50.35**	**44.13**	**49.38**
Qwen3.5-35B-A3B	44.92	44.92	41.21	43.68

Table A9: Strict accuracy (%) under different answer backbones. Each cell uses all 1,269 questions as the denominator; missing or unscored predictions count as incorrect. Only answer generation is replayed, while each method's original evidence input remains fixed.

	Strict answer accuracy			Complete access		
Method	$K = 4$	$K = 8$	$K = 10^{\dagger}$	$K = 4$	$K = 8$	$K = 10^{\dagger}$
RAG-Dense	**45.39**	34.44	34.44	76.75	89.36	89.36
RAG-Hybrid	**32.70**	30.18	30.18	48.86	72.89	72.89
A-Mem	**49.33**	38.69	38.70	76.42	90.69	90.70
Mem0-Agentic	34.20	36.33	**39.87**	–	–	–
Mem0-OSS	31.52	34.28	**36.80**	–	–	–
MuRAG	31.99	**45.63**	**45.63**	34.52	73.21	73.21

Table A10: Top-K results on all 1,269 questions (%). Complete access uses exact turn-level provenance and is not reported for Mem0's source-session proxy. †The $K = 10$ column reports the current main-experiment aggregates.

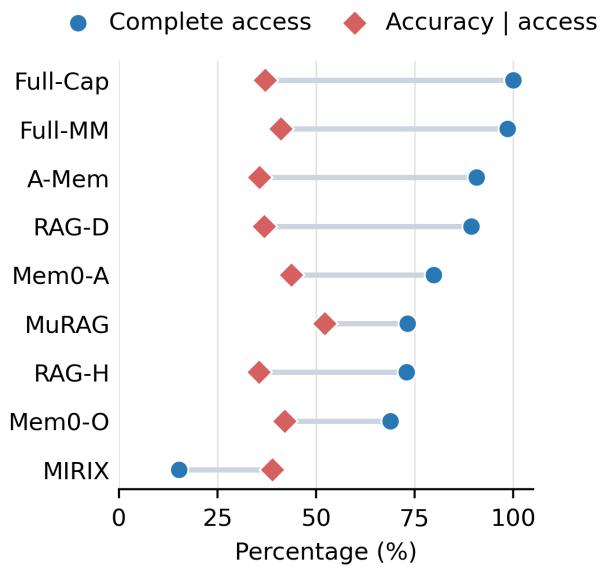

Figure A3: Complete evidence access and conditional answer accuracy. The horizontal separation between paired markers shows the post-access use gap.

assistant descriptions. Both cover all 1,269 questions.

Text-Only is 12.53 points below caption context and 17.02 points below native multimodal context. Chart accuracy is 0% under Text-Only, compared with 55.71% for captions and 32.86% for native media. Video rises from 16.67% Text-Only to 32.50% native multimodal. The nonzero Text-Only score is expected because some management questions use high-level textual signals and because refusal does not require reading a missing value. We therefore use task- and modality- specific controls instead of claiming that every single example is impossible from all text.

F.8 Context-Length Analysis

The matched 4K/8K/12K study uses Qwen3.5-27B for all three methods and should be compared only within this group. RAG-Dense declines from 59.09% to 54.71%, and Mem0-OSS declines from 55.10% to 51.02%. MuRAG is nonmonotonic, dropping at 8K and recovering at 12K. The result indicates that increasing distractor history does not automatically improve a system even when the context window can technically hold it.

The decline is not solely a retrieval-count effect. Longer histories increase the number of semantically related but answer-irrelevant sessions, and reasoning tasks require the model to compose several retrieved fragments. In the detailed breakdown, degradation is larger for reasoning than for simple retrieval in the text RAG condition. MuRAG's instability further suggests that media selection and cross-modal fusion introduce variance beyond token count.

F.9 Video Extension

All standard systems remain below 34% on the 240 video questions. MuRAG is highest at 33.75%, followed by Full-Context Multimodal at 32.50%. Caption- based and explicit-memory systems lose fine-grained action order during video compression. Native input consistently helps in paired designs: Full-Context Multimodal exceeds Full-Context Caption by 8.33 points, MuRAG exceeds RAG-Dense by 19.17 points, and native oracle exceeds caption oracle by 11.25 points.

The MIRIX result is diagnostically useful: despite a native multimodal memory design, it reaches 7.92% on this extension. Native access is therefore not equivalent to preserving

Method	Image (828)	Chart (70)	PDF (55)	Video (240)
MuRAG	51.0	75.7	32.7	**33.8**
Full-Context Multimodal	47.1	32.9	16.4	32.5
Mem0-Agentic	**51.1**	38.6	14.5	14.2
Mem0-OSS	44.0	58.6	9.1	17.1
Full-Context Caption	41.7	55.7	23.6	24.2
A-Mem	41.5	**82.9**	**36.4**	20.8
MIRIX	39.4	10.0	12.7	7.9
RAG-Dense	38.2	80.0	30.9	14.6
RAG-Hybrid	32.5	75.7	30.9	13.8

Table A11: Accuracy (%) by primary evidence modality. The 76 text or missing-media controls are omitted because they do not define a comparable native-media category.

Method	Any-hit/access	Complete access	Overall accuracy	Accuracy | complete
MuRAG	94.33	73.21	45.63	52.21
Full-Context Multimodal	100.00	98.55	41.53	41.11
Mem0-Agentic	97.32	79.83	39.87	43.73
Mem0-OSS	96.38	68.87	36.80	41.99
Full-Context Caption	100.00	100.00	37.04	37.04
A-Mem	99.61	90.70	38.70	35.62
MIRIX	44.02	15.26	30.50	38.95
RAG-Dense	99.45	89.36	34.44	36.86
RAG-Hybrid	95.04	72.89	30.18	35.46

Table A12: Evidence access and answer accuracy (%) at $K = 10$. RAG and A-Mem use exact provenance; Mem0 and MIRIX use source-session proxies; full-context rows measure delivered exposure. These provenance types are disclosed rather than treated as interchangeable retrieval recall.

Oracle input	Main	Video	All
Evidence-only Caption	67.44	35.00	61.31
Evidence-only Multimodal	63.07	46.25	59.89

Table A13: Oracle-evidence accuracy (%). Gold evidence selection is given; the answer must still interpret and integrate its content.

the event in an inspectable memory or selecting it at query time. More generally, video errors frequently involve adjacent actions, event counts, or intent cues that are omitted by a static summary.

F.10 Statistical Uncertainty and Significance Protocol

For a strict score $\hat{p} = c/N$, we report a two-sided 95% Wilson interval:

$$C = \frac{\hat{p} + z^2/(2N)}{1 + z^2/N},$$
$$R = \frac{z\sqrt{\hat{p}(1-\hat{p})/N + z^2/(4N^2)}}{1 + z^2/N}, \qquad \text{(A7)}$$
$$\mathrm{CI}_{95} = [C - R,\, C + R], \qquad z = 1.96.$$

Table A17 gives intervals for all main configurations. These intervals quantify marginal uncertainty; overlapping or nonoverlapping marginal intervals are not used as a substitute for a paired significance test.

For pairwise comparison, the released analysis script joins predictions by question ID and constructs the discordant counts n_{01} and n_{10}. It uses an exact two-sided McNemar test when $n_{01} + n_{10}$ is small and the continuity-corrected statistic otherwise. A paired bootstrap over trajectories is recommended as a robustness check because two questions from one trajectory share history. Family-wise claims across many method pairs should use Holm correction. We do not infer pairwise significance from aggregate accuracies alone.

F.11 Additional System Comparisons

The full benchmark exposes complementary method behavior. MuRAG and Mem0-A jointly answer 323 questions correctly; MuRAG uniquely answers 256 and Mem0-A uniquely answers 183. By contrast, RAG-Dense and RAG-Hybrid agree in outcome on 87.5% of questions, indicating that hybrid lexical retrieval changes fewer decisions than the difference between retrieval and explicit memory. Mem0-OSS and Mem0-Agentic agree on 77.4%. These overlaps show that the benchmark is not measuring one homogeneous source of difficulty and motivate ensemble or capability-adaptive memory policies.

G Qualitative Analysis

Appendix Figures A5–A8 visualize four representative cases that complement the aggregate results. They expose differences in final-choice tracking, temporal ordering, spreadsheet-grounded numerical comparison, and state updating. Each figure retains the original question and system

Input	Overall	Retrieval	Reasoning	Management	Image	Video
Question-Only	6.15	0.00	0.00	19.21	1.69	8.33
Text-Only	24.51	11.56	22.97	41.63	29.83	16.67
Full-Context Caption	37.04	40.16	26.76	42.61	41.67	24.17
Full-Context Multimodal	41.53	39.35	34.32	50.74	47.10	32.50

Table A14: Shortcut and full-context controls (accuracy, %). Question-Only's 78 correct answers all come from refusal-to-answer, reflecting a default refusal strategy rather than retrieval or reasoning leakage.

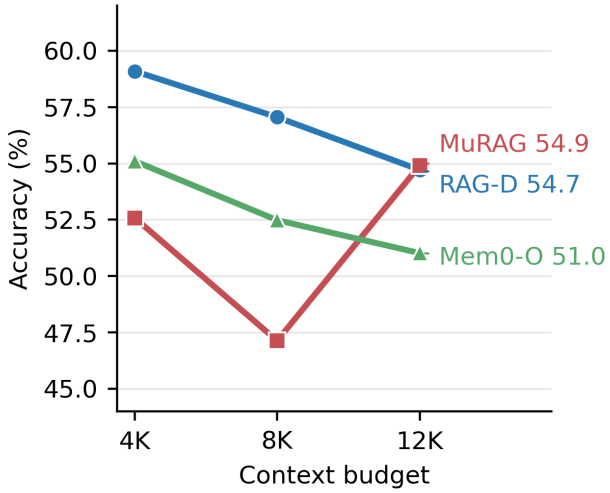

Figure A4: Matched context-length study on 1,029 image/document questions. All points use the Qwen3.5-27B answer backbone and fixed evidence.

Method	4K	8K	12K	$\Delta_{4\to12}$
RAG-Dense	59.09	57.05	54.71	−4.38
MuRAG	52.58	47.13	54.91	+2.33
Mem0-OSS	55.10	52.48	51.02	−4.08

Table A15: Accuracy (%) under matched context budgets.

outputs so that the source of the framework disagreement remains inspectable.

G.1 Successful Cross-Session Evidence Integration

In a cross-modal relation case, the question asks, "Based on this reporting goal, which chart should I cite the most, and why?" Three sessions contain charts about broadband access, public health, and unrelated regional trends. Full-Context Multimodal identifies the broadband-access chart and explains that it directly demonstrates improvement in digital access. The prediction is correct because it links a textual reporting goal to the appropriate chart, rejects visually similar distractors, and states the relation rather than only naming the media.

A successful RAG summary case asks whether the user has a healthy dietary habit. RAG-Dense retrieves six food

Method	Video accuracy
MuRAG	33.75
Full-Context Multimodal	32.50
Full-Context Caption	24.17
A-Mem	20.83
Mem0-OSS	17.08
RAG-Dense	14.58
Mem0-Agentic	14.17
RAG-Hybrid	13.75
MIRIX	7.92

Table A16: Strict accuracy (%) on the 240-question video extension.

Method	Correct / 1,269	Wilson 95% CI
MuRAG	579	[42.90, 48.38]
Full-MM	527	[38.85, 44.26]
Mem0-A	506	[37.21, 42.59]
A-Mem	491	[36.05, 41.40]
Full-Cap	470	[34.42, 39.73]
Mem0-O	467	[34.19, 39.49]
RAG-Dense	437	[31.87, 37.09]
MIRIX	387	[28.03, 33.09]
RAG-Hybrid	383	[27.72, 32.76]

Table A17: Strict accuracy counts and Wilson intervals (percent).

and calorie records at ranks one through six and answers affirmatively with a concise reference to the repeated pattern. No single meal is sufficient; success requires retrieving a distributed set and aggregating consistent evidence.

G.2 Retrieval Failure

A conflict case asks whether participating in a half marathon on 8 March 2025 is appropriate. The reference answer is no because an earlier physician's instruction requires two weeks of rest over that date. RAG-Dense retrieves semantically similar exercise and injury sessions but misses the exact medical instruction. It answers that the decision cannot be determined. Here the failure is attributable to evidence selection: the required turn is absent from the delivered top ten, and the cautious answer is otherwise reasonable.

This case illustrates why topic similarity is insufficient for memory retrieval. Several distractors contain the same entities and activity domain, but only the dated recovery constraint determines the answer. Hybrid lexical retrieval

can also fail when paraphrases share few surface terms.

G.3 Evidence-Utilization Failure after Complete Retrieval

Another conflict case asks whether daytime hiking on 18 April 2025 is compatible with the user's schedule. The required session states that a quarterly review runs from 9:00 to 17:00 and requires online attendance and a report. RAG-Dense places this evidence at rank one, satisfying complete access, but answers that it cannot determine the schedule. The error is downstream of retrieval: the model fails to map "during the day" to the occupied interval and to combine the activity with the attendance constraint.

This pattern motivates conditional accuracy. A retrieval-only metric would mark the example successful, while end-to-end strict accuracy correctly marks it wrong. Similar failures arise when every requested answer component is present but the model uses only one fragment, resolves a relative date against the wrong session, or applies an outdated state after a later update.

G.4 Native Multimodal Reasoning Failure

In a causal-intent example, the user asks why a project proposed on 6 March 2024 can no longer continue. A later chart shows a target economic measure falling from 323.6 in 2019 to 320.66 in 2020, contradicting the proposal's assumption of maintained growth. Full-Context Multimodal receives both the query context and the required chart but answers that the user found a better project. The response is plausible in isolation but unsupported.

The example separates media delivery from grounded use. The relevant pixels arrive at the model, yet the system must read two values, compare them, connect the trend to the proposal's premise, and resist a generic explanation. This multi-stage use problem is one reason native multimodal access does not match oracle-level correctness.

G.5 Video Temporal-Order Failure

A video question asks what happens immediately after "Pick the ends off the verdalago." The reference event is combining lemon juice, sumac, garlic, salt, and oil in a bowl. Full-Context Multimodal instead answers "Add red pepper flakes to the bowl." Both actions occur nearby in the cooking sequence, but only one is adjacent to the anchor event. Sparse frame sampling and holistic video summaries can preserve the recipe topic while losing exact event order.

This case reflects a common video error mode: the system recognizes objects and actions but selects a neighboring event. Increasing frame count can help only if memory construction and answer-time attention retain timestamps and local transitions rather than compressing the clip into an unordered activity list.

G.6 Error Taxonomy

We organize observed errors into six stages. (1) *representation loss* occurs when captioning or memory extraction omits an essential visual detail. (2) *selection failure* occurs when required evidence is absent from the delivered context. (3) *partial composition* uses one of several required fragments. (4) *temporal misresolution* applies the wrong timestamp, event order, or outdated state. (5) *unsupported completion* replaces grounded evidence with a plausible prior. (6) *answer realization* contains the correct reasoning but omits a requested component or expresses an incorrect final value.

Error stage	Diagnostic pattern
Representation loss	A required color, chart cell, or action is absent from the caption or memory although the gold media was delivered.
Selection failure	A distractor outranks the dated source, so required evidence is not delivered and CESA is false.
Partial composition	The answer uses only part of a multi-session evidence set; any-hit may be true while CESA is false or the answer is incomplete.
Temporal misresolution	The system binds a relative date to the wrong session or applies an outdated state despite complete access.
Unsupported completion	A plausible language-model prior overrides retrieved evidence, producing an unsupported answer with CESA true.
Answer realization	The correct evidence is available, but the output omits a field or contains a calculation error.

Table A18: Qualitative error taxonomy and its relation to auditable evidence access. CESA alone cannot distinguish all post-access categories.

Access traces distinguish category (2) from downstream failures, but not all remaining categories can be inferred automatically. We therefore combine provenance analysis with manual inspection. The central empirical pattern is not that retrieval is solved: complete access improves accuracy. Rather, the large post-access failure rate shows that improving retrieval alone cannot close the benchmark gap.

H Human Evaluation and Annotation Details

H.1 Annotator Roles and Workflow

Annotation uses multiple expert roles rather than a single unconstrained generation pass. Task designers define capability boundaries and negative constraints. Media curators verify source content and suitability. Sample authors reconstruct fine-grained memories, QA, and evidence from the media. Reviewers inspect completed long-context instances. One person can serve more than one role on different examples, but an example is not accepted solely on the judgment of its original author.

The workflow is create, validate, review, revise, and adjudicate. Reviewers see the media, full timestamped history, question, answer, and evidence manifest. They do not infer correctness from a caption alone. Disagreements identify a concrete issue, such as an unreadable value, second plausible answer, missing support, or unnatural carrier transition. The author revises the item, after which it undergoes another review. Ambiguous cases that cannot be repaired are removed.

H.2 Annotation Guidelines

Annotators apply the following acceptance criteria:

1. Every answer component is directly supported or follows by a specified operation from the evidence.
2. The required visual, document, or video detail is visible at release quality and is not stated verbatim in ordinary context text.
3. Evidence IDs identify a minimal sufficient set and survive injection.
4. Timestamps and temporal expressions are mutually consistent.
5. Distractors are plausible in the story but neither answer the question nor create an equally valid alternative.
6. The question is natural from the primary speaker's perspective and contains no task label or hidden metadata.
7. The answer is concise, unambiguous, and uses the correct unit, date, entity, and update state.

For refusal examples, reviewers actively search the complete history for the requested fact. For conflict and update examples, they write an explicit old/new or claim/constraint table before acceptance. For charts and PDFs, they recompute values from the rendered source. For video, they inspect the evidence window and neighboring events to verify temporal boundaries.

H.3 Review, Adjudication, and Agreement Scope

The construction process uses review and adjudication rather than collecting several independent categorical labels for every item. Consequently, we do not report an inter-annotator κ for dataset creation. Such a statistic would misrepresent the workflow because reviews produce revisions until one accepted artifact remains. We instead release validation status and evidence provenance for audit and describe the rejection criteria above.

Independent agreement is measured for the LLM judge. We sample 150 evaluated predictions for human rescoring. Humans and the automatic judge agree on 146 and disagree on four, yielding 97.33% observed agreement. The Wilson 95% interval is [93.34%, 98.96%]. We report observed agreement rather than Cohen's κ because the complete human/judge confusion matrix and class-prevalence strata are required for a meaningful κ estimate.

H.4 Evaluation Reliability

Human judge review applies the same semantic-equivalence rubric as automatic judging. Reviewers are shown the question, reference, and prediction but not the method name or retrieval score. A difference in verbosity does not imply an error; missing facts, wrong dates, contradictions, or unsupported additions do. The four disagreements are retained as calibration cases for future judge prompt revisions. Importantly, judge validation supports score reliability but does not replace item-level dataset validation.

All evaluation artifacts retain a question ID, prediction, reference hash, judge label, judge reason, error status, accessed-unit trace, and run-manifest digest. This permits rescoring without rerunning a memory system and makes it possible to audit a reported aggregate back to individual questions.

I Limitations and Future Extensions

I.1 Dataset and Modality Limitations

Although the benchmark covers images, charts, PDFs, and videos, it does not cover audio-only memories, live sensor streams, interactive web pages, or 3D environments. Videos total 7.52 hours and emphasize event understanding rather than feature-length narratives. PDF evaluation is affected by page rendering and payload limits, and 22 long-document oracle cases lack a verifiable full page-level native condition. The primary evidence buckets simplify questions that can in practice require more than one modality.

Carrier histories are realistic but constructed: expert-authored atomic tasks are injected into adapted LoCoMo/MMDU conversations. This offers controlled evidence and distractors but does not reproduce all distributional properties of naturally accumulated personal memory. The benchmark is English after a translation and review stage and does not test multilingual memory. Future versions should add multilingual trajectories, audio, longer videos, dynamic documents, and naturally collected longitudinal interactions with appropriate consent.

I.2 Evaluation and Provenance Limitations

Exact turn-level provenance is not available from every third-party framework. Source-session proxies for Mem0 and MIRIX can undercount a memory that preserves the fact without retaining a parseable timestamp, or over-credit a compressed memory whose source session contains more than the gold turn. We therefore separate proxy CESA from exact CESA, but stronger framework APIs that emit source spans would improve comparability.

Complete access also does not prove that a model attended to the evidence. It means that the required unit was delivered in the auditable answer context. Conversely, a semantically equivalent memory without recoverable provenance may answer correctly while appearing inaccessible. Future protocols can pair provenance with causal interventions, such as removing one gold fragment, counterfactually changing a value, or requiring evidence citations before the answer.

I.3 Model and Infrastructure Limitations

The main comparison uses one answer model to isolate memory architecture, but conclusions may change with stronger or differently trained multimodal models. Provider-hosted APIs can change serving kernels, templates, quantization, or rate limits without changing a public model name. Run manifests reduce this risk but cannot fully reconstruct a closed service. Captioning also introduces a common model-dependent bottleneck for text-only memory systems.

Strict accuracy compresses partial progress into a binary label. This is appropriate for the benchmark's answer requirements but can hide near-miss reasoning. Future work can add structured component scores, calibrated abstention metrics, retrieval latency, storage cost, token cost, and memory construction time. Repeated runs and broader backbone sweeps would separate architecture effects from model variance more fully.

I.4 Future Benchmark Extensions

Four extensions are especially promising. First, a causal evidence-use split could perturb retrieved values while keeping topics constant, directly testing whether the answer follows memory rather than priors. Second, an interactive protocol could ask systems to request missing media or clarification. Third, privacy and deletion tasks could verify whether a framework removes specified facts from all derived memories and indexes. Fourth, a continually growing stream could measure maintenance cost and catastrophic interference over months of simulated interaction.

The task taxonomy can also support capability-adaptive systems. Since no method dominates all dimensions, an agent may route dense documents to faithful caption retrieval, videos to native multimodal retrieval, and update-sensitive questions to explicit memory with provenance. Such routing should be evaluated under a fixed compute budget to avoid trading accuracy for unbounded model calls.

J Ethics, Responsible Use, and AI-Assistance Disclosure

J.1 Privacy and Media Provenance

The release uses media from open-source datasets and public sources under their applicable licenses or redistribution conditions. Package records retain source provenance without exposing author identities or machine-local paths. Public IDs are reindexed within each task, and auxiliary construction fields, credentials, service identifiers, and personal filesystem information are removed. The conversations are benchmark scenarios and should not be treated as factual profiles of real individuals.

Dataset users remain responsible for complying with source-media licenses. Where redistribution is restricted, the release should provide retrieval metadata or derived features according to the source terms rather than repackage the original asset. Sensitive-value refusal examples use synthetic conversation content and are intended to evaluate evidence calibration, not to encourage collection of real credentials.

J.2 Potential Risks and Intended Use

The benchmark is intended for research on long-term agent memory, multimodal retrieval, evidence-grounded reasoning, and memory management. It is not a certification that a system is safe for medical, financial, legal, or personal data deployment. A high score does not establish privacy compliance, reliable forgetting, or resistance to adversarial media. Models can also reproduce biases from source datasets and carrier conversations.

Evaluation outputs may reveal weaknesses in a framework, but they should not be used to infer abilities of an individual user or annotator. We recommend reporting model versions, media profiles, failure counts, and provenance type alongside scores to prevent misleading comparisons.

J.3 Annotation Labor

Human experts perform media inspection, sample reconstruction, evidence annotation, translation review, and final quality control. The workflow is designed to reserve human effort for factual and semantic decisions rather than bulk text generation. Any public release should document compensation and working conditions in the final ethics statement when institutional reporting is available, without disclosing identities during anonymous review.

J.4 AI-Assistance Disclosure

Language models assist candidate-event ideation, storyline abstraction, carrier-context adaptation, translation support, captioning, and automatic answer judging. Candidate concrete memories, questions, answers, and evidence labels generated during ideation are discarded. Human experts reconstruct and approve the final fine-grained task content against real media. Deterministic scripts perform injection, ID mapping, length pruning, validation, and metric calculation. No model is treated as the sole authority for benchmark ground truth.

Figure A5: Cross-session final-choice reasoning. Only Mem0-O among the shown systems resolves both the birthday date and the final gift, while the other methods either select an earlier gift or fail to recover the required cross-session evidence.

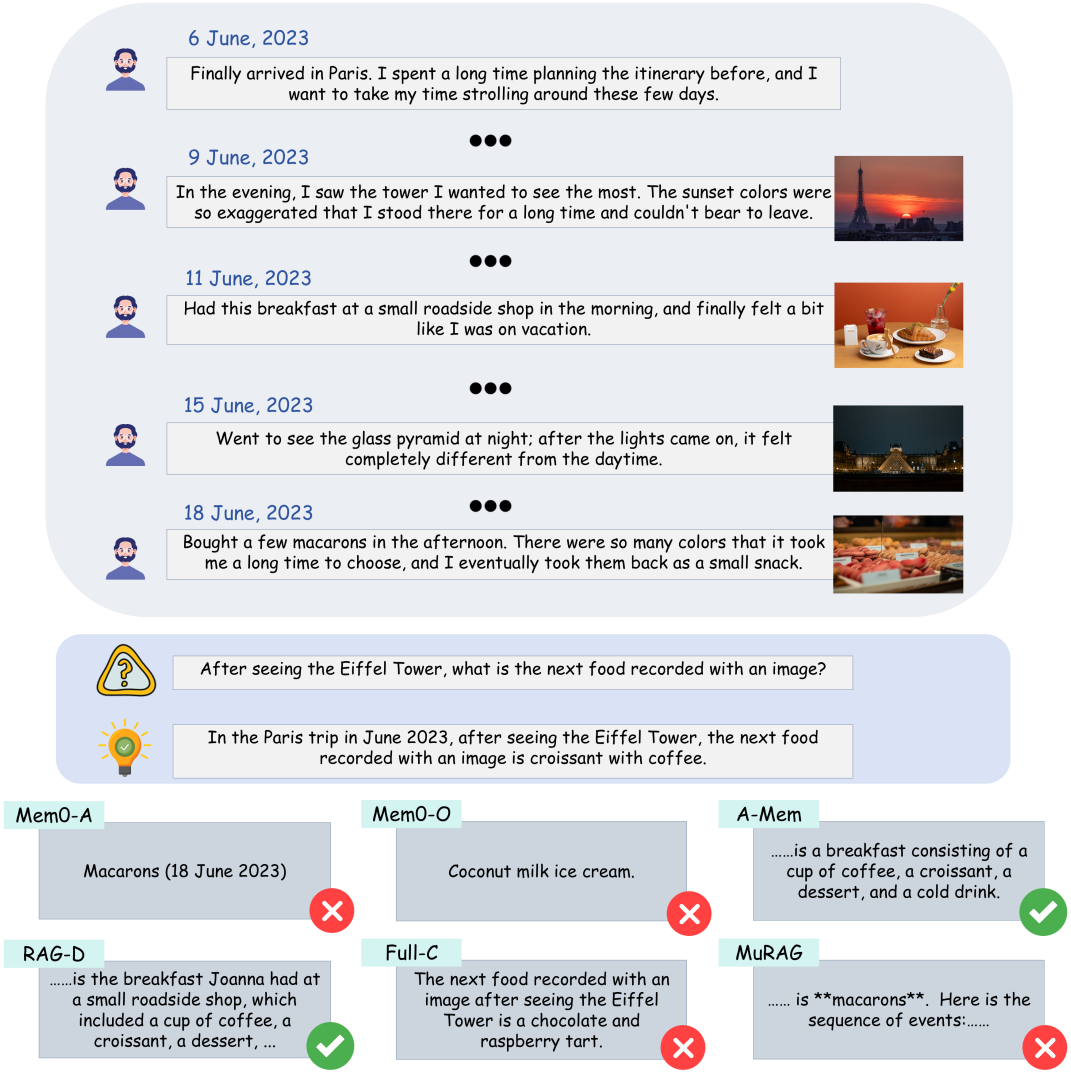

Figure A6: Temporal ordering with semantically related distractors. A-Mem and RAG-D recover the event immediately following the Eiffel Tower visit, whereas the other systems select later or unrelated food records.

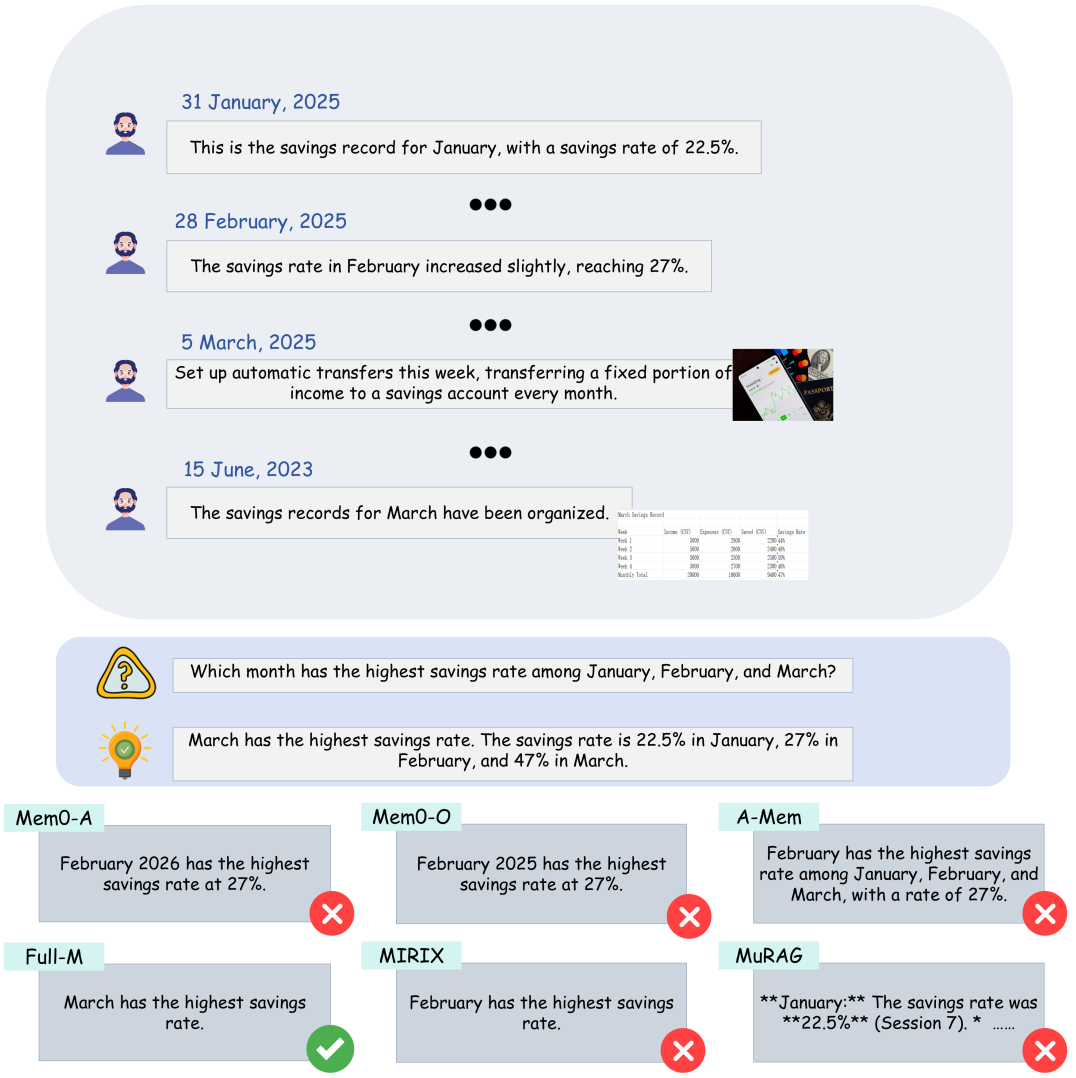

Figure A7: Spreadsheet-grounded numerical comparison. The frameworks receive monthly savings records in different forms but disagree on which month has the highest savings rate, illustrating differences in extracting and comparing fine-grained numerical evidence.

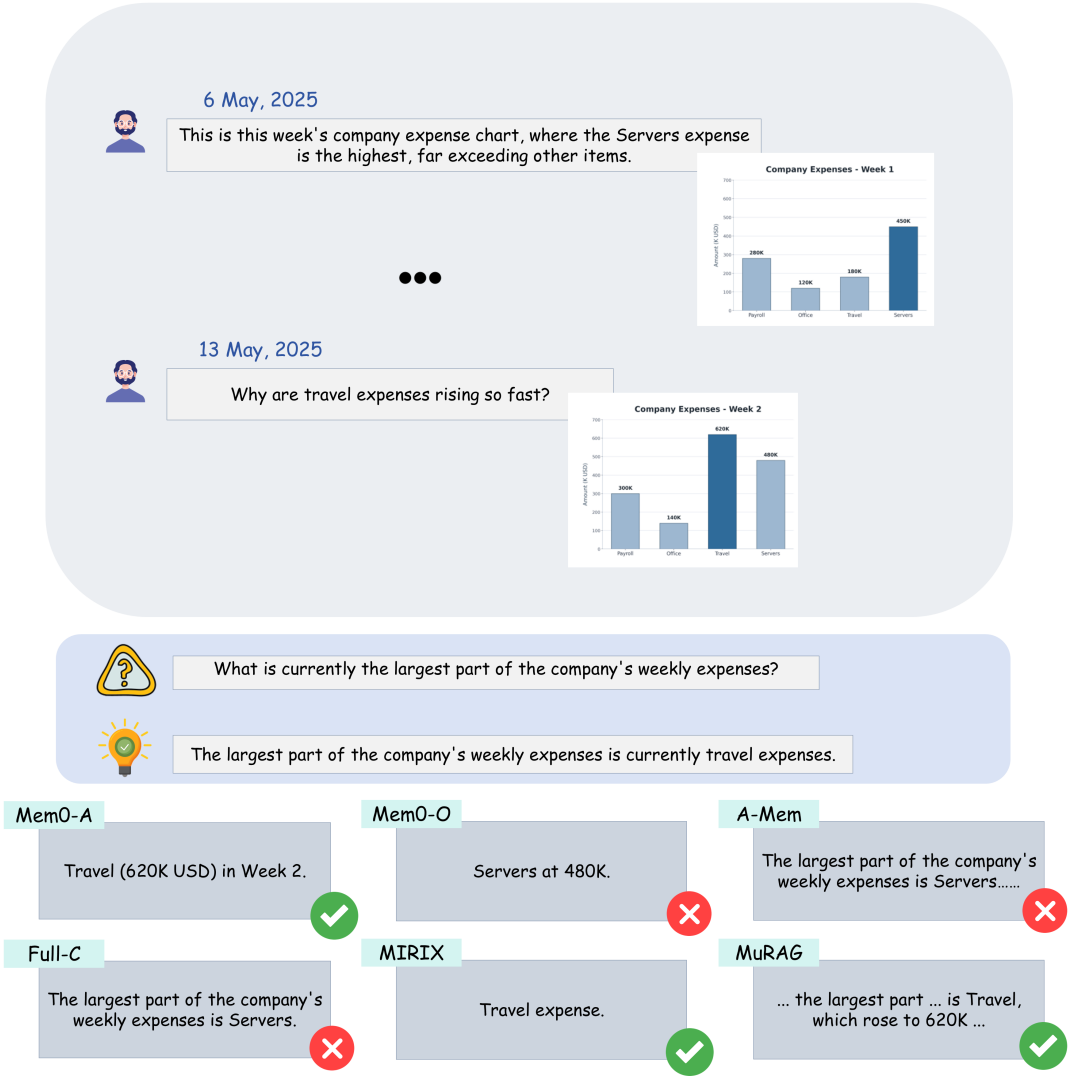

Figure A8: Updating an earlier state from a newer chart. The systems receive an older Week 1 state and a newer Week 2 state but disagree on whether Travel supersedes Servers as the current largest expense.